

Multihead self-attention across cortical and subcortical circuits

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Abstract

Both biological cortico-thalamic networks and artificial transformer networks use canonical computations to perform a wide range of cognitive tasks. We show that the structure of cortico-thalamic circuits is well suited to realize a computation analogous to multihead self-attention, the main algorithmic innovation of transformer networks. Layer 2/3 and layer 5 pyramidal cells of the cortex are assigned distinct computational roles: L2/3 pyramidal cells maintain a recurrent key-value memory, and L5 pyramidal cells encode the memory retrieved by the current query. The computation matches the fine-grained structure of core and matrix projections from the thalamus to multiple cortical areas. While a single cortical area represents one head of attention, the full thalamo-cortico-thalamic network implements a multihead linear self-attention block. Cross-attention provides the context information from the key-value memory in one cortical area to a query in another cortical area. Such lateral cortico-cortical interactions may cross-modulate the motor stream by a sensory-cognitive stream, implementing an encoder-decoder architecture. We further show how an additional discounted future reward signal can modulate the L2/3 memory, and hence the context-dependent value representation in L5. Last not least, gradient-based synaptic plasticity runs fully online, implemented in neuronal terms, without need to propagate errors back through time. This becomes possible if the same L2/3 memory neurons that modulate L5 activity also modulate L5 synaptic plasticity.

Introduction

Transformer networks (Vaswani et al. 2017) have taken the world by storm, notably revolutionizing natural language processing. The high performance of this network architecture is not restricted to language, but extends to any domain where inputs can be expressed as sequences of ‘tokens’, including vision (Dosovitskiy et al. 2021) and potentially abstract reasoning (Chollet 2024). At the hearth of transformer networks is the self-attention mechanism, a scalable and parallelizable computation that can handle long-range dependencies when processing sequences. Searching for a possible cortical implementation of such self-attention mechanisms may give hints on how cortex may reach its cognitive power.

Self-attention is based on the computation of three different quantities by linear projections of the same sequence elements (tokens): queries, keys, and values. An intuition for the different roles of these quantities is best given in terms of a soft dictionary: each key indexes a specific dictionary item (a ‘value’), and upon a query, a mixture of values is returned. Each value of this mixture is weighted by the similarity between the query and the key corresponding to that value. The initial interpretation of self-attention was as a *parallel* computation over all tokens in a temporal context window (Vaswani et al. 2017). In this interpretation, current and past tokens each compute their own key, query, and value. Each query is compared to all keys forming a multiplicative attention mask over values, and the resulting weighted sum for each token forms the output. A formally equivalent formulation (see Methods) is as a key-value *recurrent* memory system with retrieval by the current query (Kohonen 1972; Katharopoulos, Vyas, Pappas, and Fleuret 2020; Sun et al. 2023; Gershman, Fiete, and Irie 2025). In this interpretation, only the current token computes a key, a query, and a value. The current key-value pair is integrated into a recurrent memory, and the current query retrieves from the recurrent memory a weighted

sum of past values based on the similarity between the associated keys in the memory and the new query. In a *multihead* self-attention block, this mechanism is repeated in parallel across multiple independent attention heads, each computing different keys, queries, and values, thus attending to different aspects of the tokens. To form the output of a multihead self-attention block, the results of all heads are added together with output weights specific to each head (Elhage et al. 2021).

The neocortex of mammals is also built on a parallelizable canonical computational motif (Douglas and Martin 2004; Harris and Shepherd 2015; Powell, Hein, Kong, Elpelt, Mulholland, Kaschube, and Smith 2024; Meyer, Martynnek, Kastner, Livingstone, and Arcaro 2025) repeated both in parallel and in series (Felleman and Van Essen 1991; D’Souza, Wang, Ji, Meier, Kennedy, Knoblauch, and Burkhalter 2022). Concerted efforts have recently greatly advanced the data available on the local components of this canonical cortical circuit (Staiger and Petersen 2021; Sievers et al. 2024; The MICrONS consortium 2025), but a computational interpretation with demonstrably high performance on hard tasks is missing. Moreover, there is a growing interest in computational interpretations of the crucial interactions between the cerebral cortex and the thalamus (Suzuki, Pennartz, and Aru 2023; Furutachi, Franklin, Aldea, Mrcic-Flogel, and Hofer 2024; Sherman and Usrey 2024b; McKinnon, Mo, and Sherman 2025; Whyte, Müller, Aru, Larkum, John, Munn, and Shine 2025).

Previous work on biologically-plausible neural implementations of self-attention have focused on a single head, and aim to model either the hippocampal formation (Whittington, Warren, and Behrens 2022) or the role of glial cells (Kozachkov, Kastanenko, and Krotov 2023). Claims of biological plausibility have also been made on the basis of the equivalence between self-attention and the update in modern continuous Hopfield networks (Ramsauer et al. 2021; Krotov and Hopfield 2021; Hoover et al. 2023). Here we propose that the structure of cortico-thalamic circuits, cell types, pathways and interactions, are well suited to implement a computation akin to multihead self- and cross-attention. Specifically, we base ourselves on the formulation of linear self-attention as a recurrent key-value memory system (Kohonen 1972; Katharopoulos, Vyas, Pappas, and Fleuret 2020; Sun et al. 2023; Gershman, Fiete, and Irie 2025). We show how cortical layer 2/3 pyramidal neurons could form a recurrent key-value memory integrating thalamic inputs while layer 5 pyramidal neurons could perform retrieval by the current query.

We do not claim that cortico-thalamic circuits strictly implement the same computation as transformer networks, but rather, that the reported structural analogies might be illuminating for a mechanistic understanding of cortical computation and, eventually, the cognitive ability of the brain. Given the already well-known computational power of transformer circuits, we focus on the layout of the theory with a comprehensive mapping to cortical microcircuits and macroscopic structures. The mapping includes a neuronal implementation of gradient-based learning that runs purely online, without the need of backpropagation through time (BPTT). Various task-related observations, such as cortical salience signals related to prospective rewards or punishments, naturally extend the cortical transformer framework and include subcortical structures, with potential applications back to the artificial counterparts.

Results

Multihead linear self-attention

In response to an input token $\mathbf{x}_t \in \mathbb{R}^d$ at time t (a column vector), heads with index h compute keys $\mathbf{k}_t^h = \mathbf{W}_K^h \mathbf{x}_t$ with $\mathbf{W}_K^h \in \mathbb{R}^{d_k \times d}$, and values $\mathbf{v}_t^h = \mathbf{W}_V^h \mathbf{x}_t$ with $\mathbf{W}_V^h \in \mathbb{R}^{d_v \times d}$. Head-specific memory matrices $\mathbf{M}_t^h \in \mathbb{R}^{d_v \times d_k}$ recurrently integrate an association between current keys and values, alongside those of past key-value memories. At each discrete time t memory matrices are recurrently updated by the outer product of values and keys,

$$\mathbf{M}_t^h = \mathbf{M}_{t-1}^h + \frac{1}{\tau^h} \left(\mathbf{v}_t^h \phi(\mathbf{k}_t^h)^\top - g_\ell \mathbf{M}_{t-1}^h \right), \quad (1)$$

with head-specific time constants $\tau^h \in [1, \infty)$, leak factor $g_\ell \in [0, 1]$, and ϕ a nonlinearity applied elementwise. To capture the time scale of an input token as shortest temporal unit of sensory processing, say in auditory-visual speech processing, we can think of a time step in the order of 50–100 ms (van Wassenhove, Grant, and Poeppel 2007). The size of the memory update is proportional to the dissimilarity between the current key-value association and the already accumulated memory, or in other words, a key-value association is integrated in

proportion to its novelty. Below we will relate our delta rule to the temporal difference delta (δ_{TD}) that was shown to also shape the apical activity in layer 5 pyramidal neurons in the context of reinforcement learning (Schoenfeld et al. 2024). For $g_\ell = 1$, the update can be written as a convex combination of past memory and current input, a natural feature of conductance based synaptic inputs (Urbanczik and Senn 2014). For $\tau^h = 1$, we get back the formulation of Retentive Networks (Sun et al. 2023).

The different memory matrices $\mathbf{M}_t^h$ are retrieved with queries $\mathbf{q}_t^h = \mathbf{W}_Q^h \mathbf{x}_t$ with $\mathbf{W}_Q^h \in \mathbb{R}^{d_k \times d}$, forming mixtures of past values as output from head h at time t ,

$$\mathbf{o}_t^h = \mathbf{M}_t^h \phi(\mathbf{q}_t^h) = \frac{1}{\tau^h} \sum_{t' \leq t} \lambda_{t-t'}^h \mathbf{v}_{t'}^h \alpha_{t',t}^h, \quad (2)$$

with attention signal $\alpha_{t',t}^h = \phi(\mathbf{k}_{t'}^h)^\top \phi(\mathbf{q}_t^h)$ and temporal discount factor $\lambda_{t-t'}^h = (1 - \frac{g_\ell}{\tau^h})^{t-t'}$. A context-aware token $\mathbf{y}_t \in \mathbb{R}^d$ is then constructed by summing the retrieved memories $\mathbf{o}_t^h$ across heads with head-specific output weights $\mathbf{W}_Y^h \in \mathbb{R}^{d \times d_v}$,

$$\mathbf{y}_t = \sum_h \mathbf{W}_Y^h \mathbf{o}_t^h. \quad (3)$$

For $g_\ell = 0$, eqs. 1 and 3 describe a multihead, linear, unnormalized self-attention layer in the limit of an infinite context window (see Methods for more details, and Gershman, Fiete, and Irie 2025 for an introduction to the computational concepts). This is a popular alternative to the more classical softmax multihead self-attention, with extensive testing in the recent machine learning literature (Schlag, Irie, and Schmidhuber 2021; Qin, Han, Sun, Li, Kong, Barnes, and Zhong 2022; Fu, Dao, Saab, Thomas, Rudra, and Ré 2023; Von Oswald, Niklasson, Randazzo, Sacramento, Mordvintsev, Zhmoginov, and Vladymyrov 2023; Yang, Wang, Zhang, Shen, and Kim 2024).

Key-value memories in cortical layer 2/3 pyramidal neurons

To map eqs. 1 and 2 to neural circuits, we first focus on a single head h . We propose that the time-dependent memory matrix $\mathbf{M}_t^h$ is encoded by a 2-dimensional cortical sheet of $d_v \times d_k$ ensembles of layer 2/3 pyramidal neurons. A pyramidal ensemble as defined here is a tightly coupled group of pyramidal cells encoding *one* scalar quantity (i.e. one component $m_{t,ij}^h$ of $\mathbf{M}_t^h$). Every layer 2/3 pyramidal ensemble is then tasked with integrating over time the product of a specific component of the value vectors with a specific component of the key vectors (see eq. 1). The values $\mathbf{v}_t^h$ are postulated to be integrated in basal dendrites via basal synapses with weights $\mathbf{W}_V^h$, while the keys $\mathbf{k}_t^h$ are integrated in the apical dendrites via apical synapses with weights $\mathbf{W}_K^h$ (fig. 1 a). The somatic activity of an ensemble $m_{t,ij}^h$ integrates the product of the basal activity times the (putatively nonlinear) apical activity $v_{t,i}^h \phi(k_{t,j}^h)$, relying on the multiplicative apical gain modulation of the peri-somatic activity (Larkum, Senn, and Lüscher 2004).

Beside the dendritic inputs, we also consider short-range (intra-ensemble) recurrent layer 2/3 connections that integrate the somatic activity in time to form a short-term memory of the current and previous somatic activity (Holmgren, Harkany, Svennenfors, and Zilberter 2003; Funayama et al. 2015). In our context, this leads to the recurrent key-value memory with individual components represented by layer 2/3 pyramidal ensembles,

$$\text{L23}_{t,ij}^h = \overset{\text{self-recurrence}}{\text{L23}_{t-1,ij}^h} + \frac{1}{\tau^h} \left(\left[\sum_{j'}^{\text{basal}} w_{V,ij',j'}^h x_{t,j'} \right] \cdot \phi \left(\left[\sum_{i'}^{\text{apical}} w_{K,ji',i'}^h x_{t,i'} \right] \right) - g_\ell \text{L23}_{t-1,ij}^h \right), \quad (4)$$

mirroring eq. 1, where we write $\text{L23}_t^h \equiv \mathbf{M}_t^h$ for clarity of the neuronal mapping.

Query-based memory retrieval in cortical layer 5 pyramidal neurons

Each layer 2/3 ensemble is postulated to project to an ensemble of intratelencephalic (IT) layer 5 pyramidal neurons within the same vertical microcolumn (see fig. 1 a). There are therefore $d_v \times d_k$ such ensembles of layer 5 IT (L5IT) neurons, each receiving a component of the memory matrix $\mathbf{M}_t^h$ at their apical dendrites. We additionally postulate that L5IT neurons represent the query at their basal dendrites, such that a query

component is represented d_v -times distributed along the d_v -direction of the cortical sheet. Each L5IT ensemble then computes the product of its basal and apical inputs,

$$\text{L5IT}_{t,ij}^h = \phi\left(\left[\sum_{i'}^{\text{basal}} w_{Q,j,i'}^h x_{t,i'}\right]\right) \cdot \text{L23}_{t,ij}^h. \quad (5)$$

In the notation above, the L5IT neurons represent the product of the L2/3 memory matrix with the diagonal matrix of the query, $\text{L5IT}_t^h \equiv \mathbf{M}_t^h \text{diag}(\phi(\mathbf{q}_t^h))$. Finally, the activities of the L5IT pyramidal neurons are summed up in separate ensembles of layer 5 extratelencephalic tract (ET) neurons in the d_k -direction of the cortical sheet to form the head outputs

$$\text{L5ET}_{t,i}^h = \sum_j \text{L5IT}_{t,ij}^h \equiv o_{t,i}^h, \quad (6)$$

where for clarity of the neuronal mapping we introduced the notation $\text{L5ET}_{t,i}^h \equiv o_{t,i}^h$. This neuronal circuitry simply implements the matrix multiplication of eq. 2, keeping in mind that in this case the matrix $\mathbf{M}_t^h$ is encoded in the somatic activity of a neuronal population rather than more classically in synaptic weights.

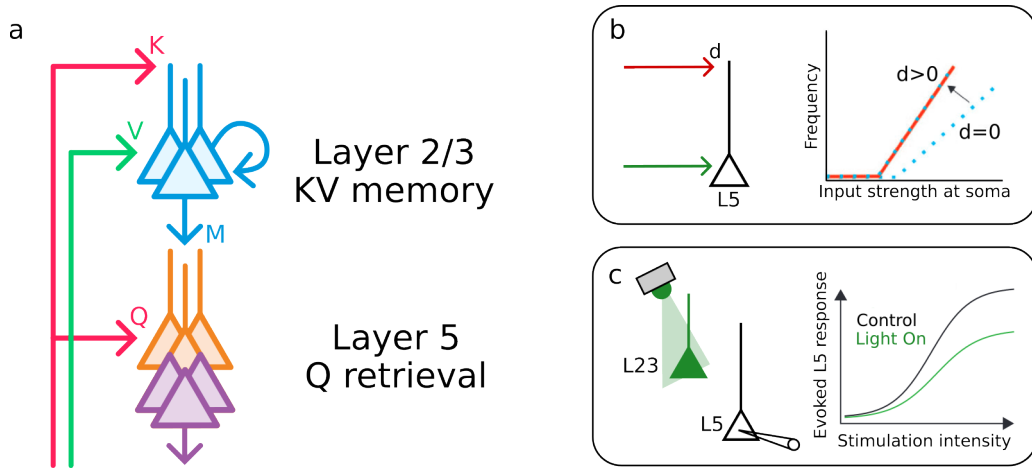


Figure 1: A cortical microcolumn of layer 2/3 and layer 5 pyramidal ensembles implement the atomic computation of attention. (a) A layer 2/3 pyramidal ensemble (blue) integrates over time pairs of keys ($\mathbf{k}$, red) and values ($\mathbf{v}$, blue). A L5IT pyramidal ensemble (orange) integrates a mixture of past key-value memories from the layer 2/3 ensemble ($\mathbf{M}$, blue, eq. 4) that modulates the current query ($\mathbf{q}$, red). The layer 5 ensemble effectively forms a sum of the attention-modulated values, with attention representing the key-query similarity. The layer 5 output is sent to the thalamus (via specific layer 5 ET (L5ET) neurons, eqs. 5 and 6, purple). (b) Apical input has a multiplicative impact on the somatic firing of layer 5 pyramidal cells. Adapted from Larkum, Senn, and Lüscher (2004). (c) Layer 2/3 pyramidal cells modulate the gain of layer 5 pyramidal cells. In this experiment layer 2/3 pyramidal cells are photo-inhibited (green line). Adapted from Quikempoix, Fayad, Boutourlinsky, Leresche, Lambert, and Bessaih (2018).

To reiterate: L2/3 pyramidal cells store key-value pairs over time, while layer 5 pyramidal cells represent the mixture of values retrieved by the current query. Both operations demand that pyramidal ensembles compute the product of their two classes of dendritic inputs, as has been shown in particular for L5 pyramidal cells (see fig. 1 b; Larkum, Senn, and Lüscher 2004). The local pathway from L2/3 to L5 pyramidal cells is established as a major component of the classical local canonical cortical circuit (Douglas and Martin 2004). Moreover, there is experimental evidence that L2/3 pyramidal cells act as controllers of the gain of L5 pyramidal cells (see fig. 1 c; Quikempoix, Fayad, Boutourlinsky, Leresche, Lambert, and Bessaih 2018).

Core and matrix thalamo-cortical projections compute keys, queries, and values

To realize eqs. 1 and 2 in the grid of $d_v \times d_k$ cortical microcolumns (see fig. 1), the inputs to these microcolumns need to be specifically distributed. According to the outer product $\mathbf{v}_t^h \phi(\mathbf{k}_t^h)^T$, the same value component $v_{t,i}^h$ is paired in the d_k -direction with all the different key components $k_{t,j}^h$ for $j = 1, \dots, d_k$. In turn, the same key component $k_{t,j}^h$ is paired in the d_v -direction with all the different value components $v_{t,i}^h$ for $i = 1, \dots, d_v$.

Anatomically, there are d_v macrocolumns, each comprising d_k microcolumns. Each value component $v_{t,i}^h$ is assigned a macrocolumn, and within this macrocolumn that value is modulated by all the different key components. Each macrocolumn (indexed by i) is also assigned an ensemble of L5ET neurons that sums up the various key-modulations of that value $v_{t,i}^h$ within its macrocolumn (see eq. 6).

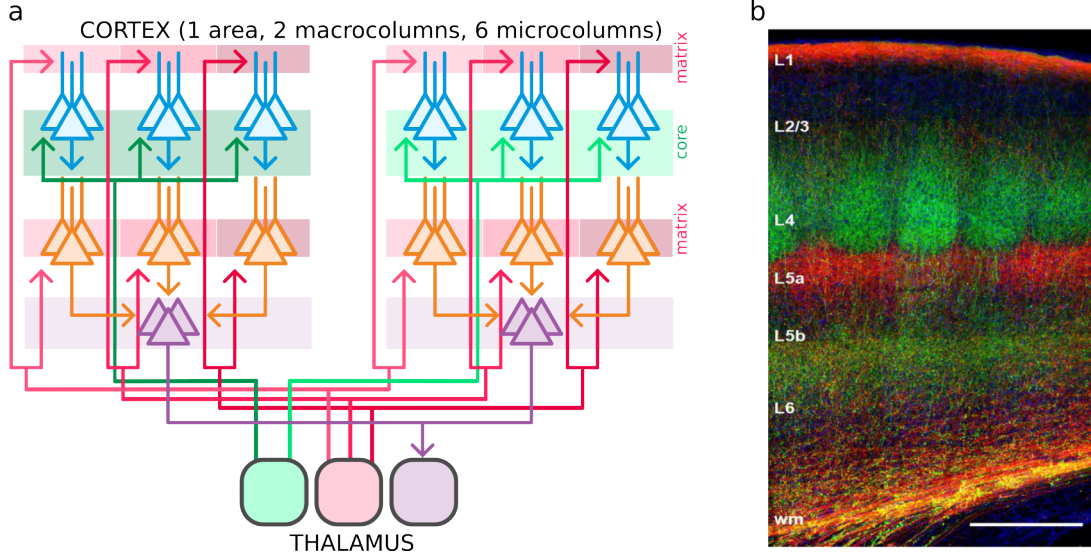


Figure 2: Core/Matrix structure of thalamo-cortical projections may compute keys, queries, and values. (a) One head (h) of linear self-attention with 2×3 cortical microcolumns, arranged in 2 macrocolumns ($d_v = 2$, $d_k = 3$). Green and red bottom squares represent a thalamic nucleus encoding the current input token $\mathbf{x}_t$ and sending core-type and matrix-type projections to the cortex, respectively. Shades of colours indicate which component of the key ($\mathbf{k}_t^h$, upper red matrix projections), value ($\mathbf{v}_t^h$, green core projections) and query vector ($\mathbf{q}_t^h$, lower red matrix projections) a particular ensemble is receiving. One vertical set of a layer 2/3 and layer 5 pyramidal ensemble act as in fig. 1. Attention-modulated values encoded in L5IT pyramidal cells (orange) are integrated additively in L5ET pyramidal cells (purple) forming the output $\mathbf{o}_t^h$ of the cortical area h . Purple projections convey this output to a higher-order thalamic nucleus (purple square) with weights $\mathbf{W}_Y^h$ (forming the activity $\mathbf{y}_t$). (b) Fluorescence imaging data for core (green) and matrix (red) thalamo-cortical projections reproduced from Staiger and Petersen (2021). Scale bar is 250 μm .

Each microcolumn within a macrocolumn receives the same value ($v_{t,i}^h$, green in fig. 2 a), and different key and query components ($k_{t,j}^h$ and $q_{t,j}^h$, red in fig. 2 a). A defining feature of macrocolumns is that core thalamo-cortical projections to microcolumns within such a macrocolumn are denser than the projections distributed over multiple macrocolumns (Mountcastle 1997; Buxhoeveden and Casanova 2002; see fig. 2 b). Therefore, value projections can be considered to be dense (contact all pyramidal ensembles in a macrocolumn), focused (project to only one macrocolumn), and target the basal trees of layer 2/3 pyramidal neurons. In contrast, key and query projections need to be sparse (contact only one ensemble in a macrocolumn), diffuse (project to all macrocolumns), and target the apical trees of layer 2/3 pyramidal neurons and the basal trees of L5IT pyramidal neurons, respectively.

This corresponds well to the structure of core and matrix thalamo-cortical projections (Jones 1998; Staiger and Petersen 2021; but see Sherman and Usrey 2024a). Core projections (green in fig. 2 b) are dense topographic projections targeting mainly cortical layers 3 and 4, and are suggested to convey values ($\mathbf{v}_t^h$). They contact the basal dendritic tree of layer 2/3 pyramidal cells either directly or relaying through layer 4 neurons, again a major component of the classical cortical microcircuit (Douglas and Martin 2004). In at least the well studied rat somatosensory cortex, the extent of core projections in layer 4 are the basis for the definition of macrocolumns or ‘barrels’ (Staiger and Petersen 2021). Matrix projections (red in fig. 2 b) are more diffuse thalamo-cortical projections targeting cortical layers 1 and 5a, and are suggested to convey keys and queries ($\mathbf{k}_t^h$ and $\mathbf{q}_t^h$). In this interpretation, first-order thalamic nuclei encode the original sequence elements or tokens ($\mathbf{x}_t$), while higher-order thalamic nuclei receive back retrieved memories from all the heads ($\mathbf{o}_t^h$) and integrate them into the final output ($\mathbf{y}_t$).

Cortical areas as attention heads

We have until now described the computation in a single attention head of a transformer network. *Multihead* self-attention entails running multiple such heads independently in parallel and integrating their results additively with head-specific output weights (see eq. 3; Elhage et al. 2021). We thus propose that the correct analogy is between one head of attention and one cortical area (see fig. 3). In doing so, we depart from classical models inspired by machine learning and do not consider a strict hierarchy of cortical areas. Rather (following e.g. Suzuki, Pennartz, and Aru 2023), at each stage of computation, a subset of cortical areas processes different aspects of the same input in parallel. For example, in the visual system the heads h are *shape*, *color*, *motion*, etc... One thalamic nucleus projects to multiple cortical areas in parallel, computing independently in each of them different keys, queries, and values from the same current sequence element. The output of each area is integrated additively in another, higher-order thalamic nucleus to which all L5ET pyramidal cells in the relevant cortical areas project. The activity of this higher-order thalamic nucleus is then the input of the next stage of processing, forming the cortical hierarchy in the classical sense.

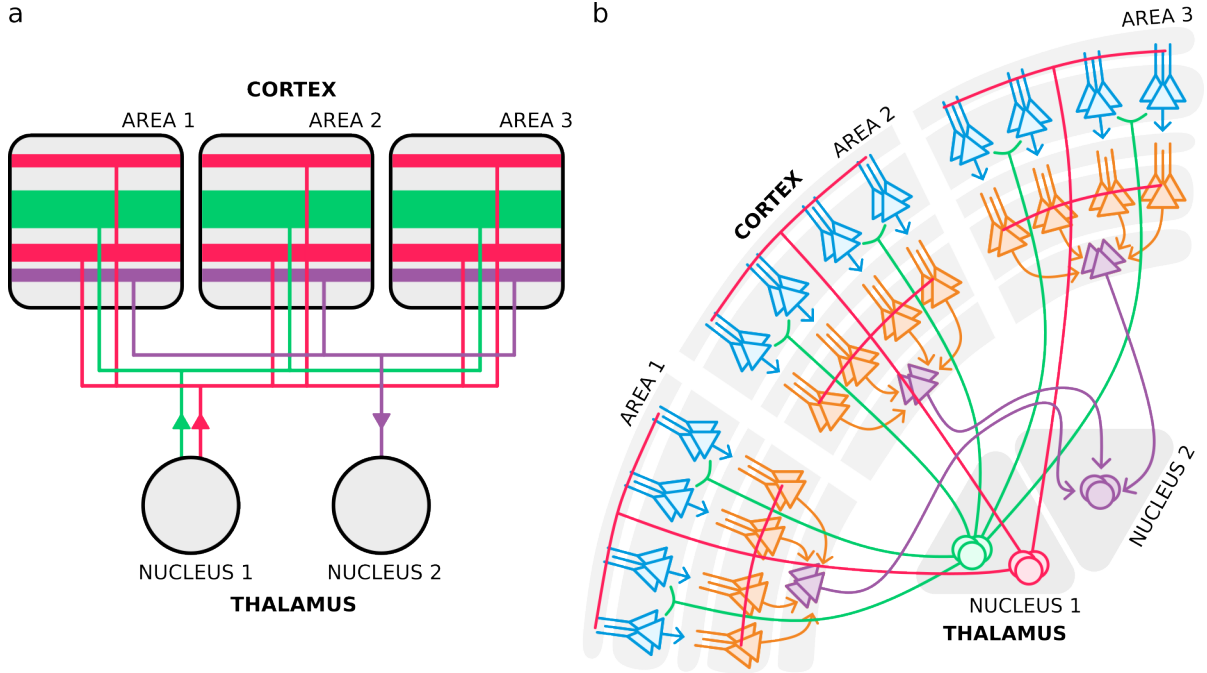


Figure 3: Cortical areas as attention heads. ‘Transcortical’ divergent and convergent pathways for multihead attention. (a) Red (matrix) thalamo-cortical projections compute keys ($\mathbf{k}_t^h$, layer 1 component) and queries ($\mathbf{q}_t^h$, layer 5 component) out of the input $\mathbf{x}_t$ from nucleus 1. Green (core) thalamo-cortical projections compute values ($\mathbf{v}_t^h$). The detailed connectivity and computation of one cortical area (square) is depicted in fig. 2. Purple cortico-thalamic projections from L5ET pyramidal cells form the output of an area/head, transporting the head-specific retrieved memories $\mathbf{M}_t^h \phi(\mathbf{q}_t^h)$, and are summed up across areas/heads (h) in the higher-order thalamic nucleus 2 (representing the activity $\mathbf{y}_t$). (b) A more biological depiction of the same concepts.

In other words, the cortico-thalamic complex forms a shallow hierarchy where cortical areas process different aspects of the input in parallel before integrating the results back in the thalamus (Mumford 1991). This general organization into divergent-convergent ‘transcortical’ pathways connecting two thalamic nuclei through a subset of cortical areas is consistent with the observation that single thalamic cells integrate information from different cortical areas (Sampathkumar, Miller-Hansen, Sherman, and Kasthuri 2021), and that L5ET pyramidal cells avoid driving back the thalamic nuclei projecting to their cortical area (Cassidy, Macias, Lagos, Ugorji, and Callaway 2025). This view, of course, does not exclude that cortical areas integrate across thalamic nuclei or other cortical areas, as we discuss next.

Thalamic and cortical cross-attention

In the following sections, we revisit the computation performed by a single head, and omit the head index h in the formalism for clarity. A natural extension of our model is to include cross- (rather than self-) attention (e.g. as already present in Vaswani et al. 2017). In that case, the current token z_t from *another* sequence $Z = \{z_1, \dots, z_t\}$ queries the key-value memory formed by the first sequence $X = \{x_1, \dots, x_t\}$. This results in head output

$$o_t^{XZ} = M_t^X \phi(q_t^Z), \quad (7)$$

where M_t^X is the key-value memory matrix formed by integrating tokens from X following eq. 1 and $q_t^Z = W_Q^Z z_t$ is a query formed by the last token from Z . The two different sequences may encode information from two different modalities. For example, the tokens x_t may encode auditory input while z_t encode visual input in the corresponding primary thalamic nuclei. In that case, such a cross-attention head is analogous in our framework to an associative (multimodal) cortical area. Multiple such areas can exist, analogous to the multiple heads of a cross-attention layer in artificial networks.

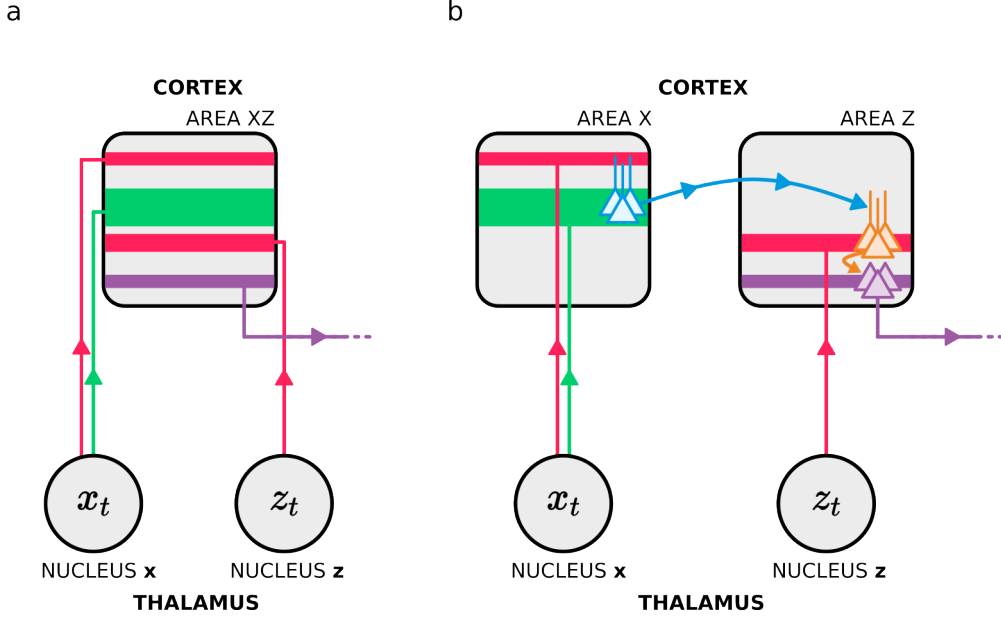


Figure 4: Two circuits for cross-attention. (a) Thalamus-driven multi-modal cross-attention in an associative cortical area. Two thalamic nuclei encode the current element of two different sequences x_t and z_t and project to a single cortical area. Projections from the nucleus encoding x_t target the superficial cortical layers and update the key-value memory. Projections from the nucleus encoding z_t target cortical layer 5 and query the L2/3 memory formed by the input x_t . The output o_t^{XZ} represents the key-value memory of modality X queried by modality Z . (b) Cortico-cortical cross-attention. Each thalamic nucleus encoding the current element for different sequences project to a different cortical area. Nucleus x updates the key-value memory in layer 2/3 of area X . Crucially, cortical area X sends to cortical area Z its key-value memory via an interareal projection (blue). In area Z , the key-value memory of area X is integrated with the query from thalamic nucleus z in L5IT pyramidal cells (orange). The output o_t^{XZ} is formed in L5ET pyramidal cells (purple) of that second area Z , which then project to another thalamic nucleus. Area X still produces its own output y_t^X via its layer 5 (not shown).

Structurally, cross-attention could be realized in multiple ways. We briefly describe two of them, both implementing eq. 7. A first realization entails thalamo-cortical projections from two different thalamic nuclei encoding respectively x_t and z_t to the same associative cortical area XZ (see fig. 4a). The thalamic nucleus encoding x_t targets both the basal and apical dendrites of layer 2/3 pyramidal neurons and updates the key-value memory M_t^X (see eqs. 1 and 4). The other thalamic nucleus encoding z_t targets the basal dendrites of L5IT pyramidal neurons in the same area and form the queries q_t^Z . A second realization of cross-attention entails cortico-cortical projections between two different cortical areas X and Z , say the primary auditory and visual cortices, driven by the two different thalamic nuclei (fig. 4b). In area X the key-value memory M_t^X is calculated in layer 2/3 based on token x_t as previously. Layer 2/3 pyramidal ensembles of area X then projects to the apical dendrites of L5IT pyramidal neurons *in area Z*. The basal dendrites of the same L5IT pyramidal neurons calculate the queries q_t^Z based on the token z_t from the other thalamic nucleus.

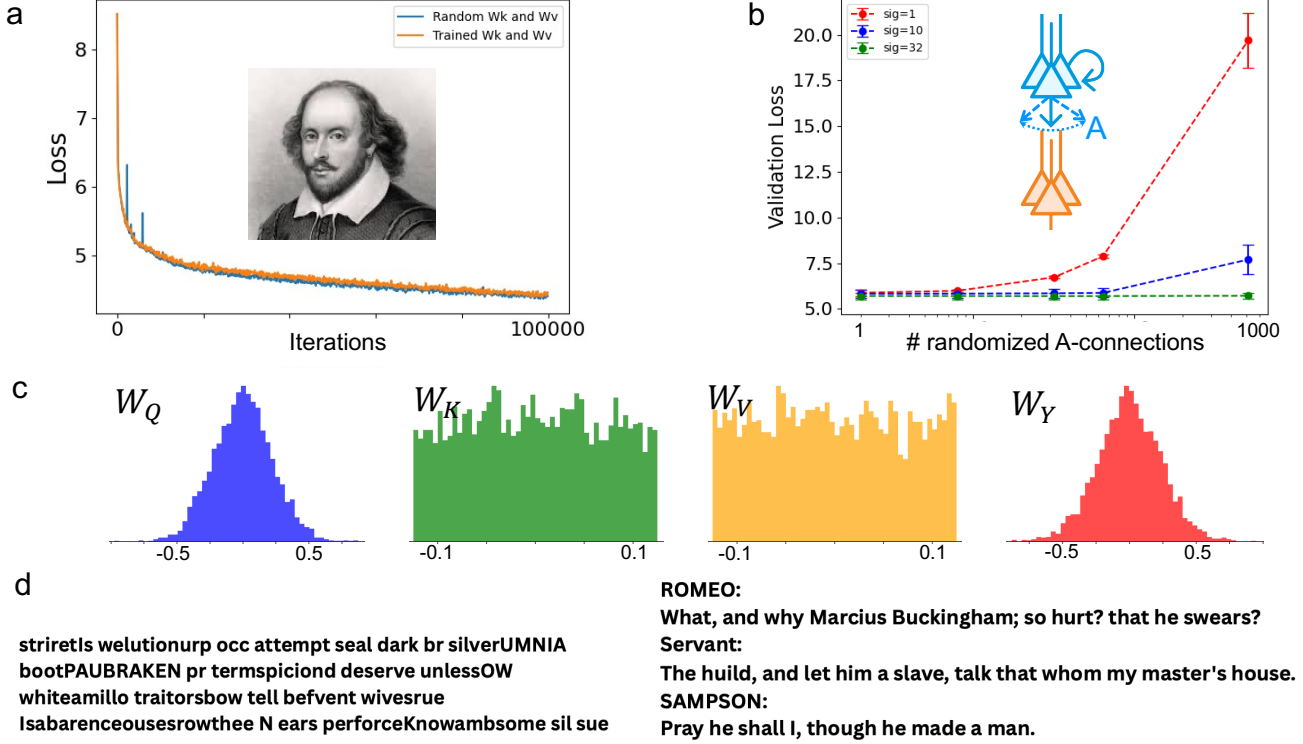


Figure 5: Tiny Shakespeare with fixed memory matrices and randomized attention connections. (a) We trained a cortical MHA network following algorithm 3 on the TinyShakespeare data set, with 4 sequential MHA blocks, each block with 4 areas ('heads'), and each area with $64^2 = 4096$ L2/3 and L5 pyramidal neurons (Methods). The same loss is achieved with memory matrices (W_K and W_V) being fixed (blue) or learned (orange). (b) In the presence of randomized connections from L2/3 to the apical dendrite of L5 neurons, the validation loss is stabilized with a broader distribution of attention weights A (red: $\sigma = 1$, blue: $\sigma = 10$, green: $\sigma = 32$, with embedding dimension $d = 64$; for the definition of A see after eq. 6). (c) Histograms of the learned query and readout synapses (W_Q and W_Y) and fixed memory synapses across the 16 cortical areas. Before learning, W_Q and W_Y histograms are as for the memory synapses. (d) Output of the network before (left) and after learning (right).

Memory-modulated sensory-cognitive-motor mappings

In area X the key-value memory M_t^X is calculated in layer 2/3 based on input x_t from the auditory kernel, say. This memory projects to the apical dendrites of layer 5 pyramidal neurons in area Z . The basal dendrites of these layer 5 pyramidal neurons calculate the queries W_Q^Z based on the input z_t from the visual kernel. These queries may then be modulated by the memory M_t^X from area X (or also from a hippocampal region), in addition to a possible memory modulation by M_t^Z from within the same area Z ,

$$z_t^{X,Z} = W_Y^Z (M_t^X + M_t^Z) \phi(W_Q^Z z_t). \quad (8)$$

The output $z_t^{X,Z}$ of area Z may project back to yet another thalamic kernel, or another cortical area (fig. 4b, e.g. via layer 4), and the same for the output y_t^X of area X (Shepherd and Yamawaki 2021). The two types of cross-attention, the multi-modal thalamo-cortical cross-attention (eq. 7) and the cortico-cortical cross-attention (eq. 8), may also be mixed.

The original transformer version in Vaswani et al. 2017 considers either self-attention (M_t^Z) or cross-attention (M_t^X in eq. 8, also shown in fig. 4b) only. Yet, the combination of self- and cross-attention, as expressed in eq. 8, seems to be an ubiquitous motif of cortical sensory, associative and motor areas. In fact, attentional modulation via top-down input to apical dendrites of cortical pyramidal neurons, implementing cross-attention, has been shown to be involved in perceptual tasks in mice and monkeys (Manita et al. 2015; Nandy, Nassi, and Reynolds 2017), as well as motor tasks (Cichon and Gan 2015; Mao, Kusefoglu, Hooks, Huber, Petreanu, and Svoboda 2011). The apical memory modulation (M_t^X) may further arise via cortico-hippocampal loop, forming a hippocampal cross-attention by episodic memories on different time scales ($\bar{\lambda}^X$, Doron et al. 2020). Apical

and basal dendrites of cortical pyramidal neurons may be simultaneously innervated from either the same cortical area (Petreanu, Mao, Sternson, and Svoboda 2009) or the same thalamic nucleus (Rah et al. 2013; Balcioglu et al. 2023), forming cortical or thalamic self-attention ($\mathbf{M}_t^Z$). The final thalamic output sums up the various self(Z)- and cross(X)-attention modulated values, $\mathbf{z}_t^+ = \sum_{X,Z} \mathbf{z}_t^{X,Z}$, with modulations originating from the different cortical or subcortical regions that each represents its own type of memory, see also eq. 3.

Given that memories $\mathbf{M}_t^X$ from many different cortical areas X modulate a specific area Z via apical gain modulation (Huang, Wu, Sansone, Ibrahim, and Fishell 2024), and that each of these areas ‘speak their own language’, the target area Z has to learn to decode these languages into its own. One solution is that each memory projects to specific L5 pyramidal neurons (via $\mathbf{W}_Q^{XZ}$) speaking the presynaptic language, and that the cross-attention modulated information is translated back to the postsynaptic language (via $\mathbf{W}_Y^{XZ}$). The cross-attention from eq. 8 then generalizes to cross-attention by a subnetwork $\mathcal{X}$ of cortical areas associated to area Z via attention matrix $\mathbf{A}^{XZ}$ telling whether area X projects to area Z ,

$$\mathbf{z}_t^{X,Z} = \sum_{X \in \mathcal{X}} \mathbf{W}_Y^{XZ} \mathbf{M}_t^X \mathbf{A}^{XZ} \phi(\mathbf{W}_Q^{XZ} \mathbf{z}_t); \quad \mathbf{z}_t^+ = \sum_Z \mathbf{z}_t^{X,Z}. \quad (9)$$

Here, $\mathbf{z}_t^+$ is the activity of a next higher-order thalamic nucleus (out of 29 nuclei, see Kumar, Beckmann, Scheffler, and Grodd 2022) that acts as a hub and integrates activity from the different cortical areas Z (Hwang, Bertolero, Liu, and D’Esposito 2017).

We hypothesize that the cortex is built of cortico-thalamic cross-attention blocks (as expressed in eq. 9) that form memory-modulated sensory-cognitive-motor mappings. More concretely, sensory inputs $\mathbf{x}_t$ are transformed into motor outputs $\mathbf{z}_t$ according to an encoder-decoder architecture (see Vaswani et al. 2017 and fig. 4b). The sensory-cognitive pathway produces encoder tokens $\mathbf{x}_t \rightarrow \mathbf{y}_t \rightarrow \mathbf{y}_t^+ \rightarrow \dots$ from the sensory areas up to higher cortical areas. The motor pathway produces decoder tokens $\mathbf{z}_t \rightarrow \mathbf{z}_t^+ \rightarrow \dots \rightarrow \mathbf{z}_{t+1}$ along a motor loop, relating the motor output at time t to a possibly stochastic motor output at time $t+1$. This motor loop is modulated via cross-attention by the key-value memories $\mathbf{M}_t^X, \mathbf{M}_t^Y \dots$, formed in the sensory stream (eq. 9). The motor loop, more specifically, is initiated by proprioceptive input encoding the motor action $\mathbf{z}_t$. The motor output $\mathbf{z}_{t+1}$, in turn, translates to the proprioceptive input at the next time step, that is complemented by the next sensory input $\mathbf{x}_{t+1}$.

Error-correcting plasticity via layer 2/3 memory, yet without BPTT

In this section we come back to self- (rather than cross-) attention and tackle synaptic plasticity for parameter learning. Transformers are often trained in supervised settings, and a full cortical implementation thus demands synaptic plasticity rules that follow the gradients of a loss function. To obtain a neuronal form of gradient-based plasticity rules, we formally derive the gradients of an autoregressive squared error loss with respect to the parameters of a multihead self-attention block (see Methods, eqs. 31 to 34).

The output synapses $\mathbf{W}_Y^h$ targeting thalamic neurons can simply be learned by the classical perceptron learning rule that directly corrects for the output error $\mathbf{e}_t = \mathbf{x}_{t+1} - \mathbf{y}_t$, with $\mathbf{y}_t$ defined by eq. 3. This gradient rule has the form of a postsynaptic (thalamic) error times the presynaptic (cortical) activity (see Methods, eq. 41),

$$\dot{\mathbf{W}}_Y^h \propto \mathbf{e}_t \mathbf{o}_t^{h\top}. \quad (10)$$

The biological interpretation is that the error $\mathbf{e}_t$ is nudging the somatic activity that in addition is driven by the dendritic input $\mathbf{o}_t^h$. But locally at the dendrite the own input can be explained away, and from the spikes propagating back into the dendrites, the synapses infer the additional somatic nudging $\mathbf{e}_t$ (Urbanczik and Senn 2014).

In the following, we focus on the linear case $\phi(x) = x$ as it allows a more straightforward neuronal implementation. Most notably, in that case, plasticity for $\mathbf{W}_{\{K,V,Q\}}$ can be expressed based on the current input token and matrix-valued decaying eligibility traces (see eqs. 36 to 40), most similar to Real Time Recurrent Learning (RTRL) rather than BackPropagation Through Time (BPTT). This gives us the capacity to learn online, while the higher computational cost entailed by matrix-valued eligibility traces is in theory offset by the high parallelism of neural hardware.

The learning of the query synapses $\mathbf{W}_Q^h$ involves the thalamic error $\mathbf{e}_t$ being backpropagated through the chain of computation,

$$\dot{\mathbf{W}}_Q^h \propto \mathbf{e}_t^{q,h} \mathbf{x}_t^\top, \quad (11)$$

with $\mathbf{e}_t^{q,h} = \mathbf{M}_t^{h\top} \mathbf{b}_t^h = \mathbf{M}_t^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t$. The error is first backpropagated from the thalamus to the cortex through weights $\mathbf{W}_Y^h$, and then through the associative memory matrix $\mathbf{M}_t^h$. We have shown in previous section how matrix multiplication is possible even if the matrix is encoded in neuronal activities rather than synaptic weights, as is the case here. The gradient plasticity for these weights are again of the form of a postsynaptic (cortical) error times the presynaptic (thalamic) activity (Methods, eq. 42). Here, our neuronal representation of the memory in neuronal activity pays out. Indeed, the eligibility trace for $\mathbf{W}_Q^h$ is simply $\mathbf{M}_t^{h\top}$, encoded in neuronal activity and thus easily able to dynamically influence plasticity. This is the main advantage of our cortical transformer implementation, as opposed to implementations inspired by Hopfield networks (Ramsauer et al. 2021; Millidge, Salvatori, Song, Lukasiewicz, and Bogacz 2022; Gershman, Fiete, and Irie 2025). In Hopfield-type implementations, the memory $\mathbf{M}_t^h$ is implemented in the *synapses* of an associative memory, not in *neurons* as done here, that can communicate their activity to modulate plasticity.

The other parameters ($\mathbf{W}_K^h$ and $\mathbf{W}_V^h$) could be learned online too (see Methods, eqs. 36 to 40), but a neuronal implementation would be more involved, as it requires the storage of new eligibility traces. Instead, we will exploit an intriguing property of key-value memory: memory capacity is maximized if the ‘memory elements’ are uniformly distributed in the ‘memory space’ (Chandra, Sharma, Chaudhuri, and Fiete 2025; Gershman, Fiete, and Irie 2025). In our setting, this implies the optimality of projecting tokens with fixed random matrices $\mathbf{W}_V^h$ and $\mathbf{W}_K^h$ when forming values $\mathbf{v}_t^h$ and keys $\mathbf{k}_t^h$, which are then integrated into the associative memory $\mathbf{M}_t^h$. No impairment in the performance is observed when these weights were fixed, as opposed to being learned by gradient descent (fig. 5 a,c). Later, we will show that biology yet found a way to learn the importance of a memory by strengthening those memories that are predictive for future reward or punishment. Given the fixed ‘memory encoding’ matrices ($\mathbf{W}_K^h$ and $\mathbf{W}_V^h$), it becomes crucial to learn the query and output weights ($\mathbf{W}_Q^h$ and $\mathbf{W}_Y^h$), respectively indicating where the appropriate memory upon input $\mathbf{x}_t$ is encoded and how the retrieved memory has to be translated back from its random encoding. These two weights can be interpreted as performing a ‘readout’ of the associative memory.

If we consider serially connected attention layers, the output error in the last layer has to be iteratively backpropagated across the layers. The formal gradients (see eqs. 48 to 50) indicates that this involves the combination of an instantaneous error backpropagation through $\mathbf{W}_Q^{h\top}$, namely $\sum_h \mathbf{W}_Q^{h\top} \mathbf{e}_t^{q,h}$, and a slower (decaying sum) component through $\mathbf{W}_V^{h\top}$ and $\mathbf{W}_K^{h\top}$.

Biological implementation of the gradient plasticity

The first step of error backpropagation from the thalamus to the cortex, computing $\mathbf{b}_t^h = \mathbf{W}_Y^{h\top} \mathbf{e}_t$, is mediated by the transpose of a synaptic weight matrix. Hence this matrix transpose can be learned by a correlation-based plasticity, as previously shown (Max, Kriener, Pineda García, Nowotny, Jaras, Senn, and Petrovici 2024). However, implementing eq. 11 further requires that these backpropagated ‘pass through’ the transpose of the temporally changing matrix $\mathbf{M}_t^h$, encoded in neuronal activity rather than synaptic weights. The cortical error $\mathbf{e}_t^{q,h}$ can therefore not be constructed from $\mathbf{b}_t^h$ via synaptic projections with transposed weights. Instead, we postulate that $\mathbf{e}_t^{q,h}$ is represented by dedicated error neurons that are modulated by the layer 2/3 memory via apical gain modulation.

In analogy to the layer 5 signal representation (eqs. 5 and 6), we postulate an explicit representation of errors in cortical layer 6, specifically, in the somatic activity of layer 6 pyramidal neurons (see e.g. English, Nejad, Sommerfelt, Yanik, and Von der Behrens 2023 for experimental evidence of Bayesian surprise encoding in the spiking activity of layer 6 neurons). We suppose that the thalamic errors $\mathbf{e}_t$ backpropagated to the cortex with weights $\mathbf{W}_Y^{h\top}$ are integrated in the basal dendrites of layer 6 IT (L6IT) pyramidal neurons. Similarly to L5IT neurons, this basal activity is gain-modulated by the corresponding apical input from layer 2/3 pyramidal neurons of the same microcolumn, such that the somatic activity is

$$\text{L6IT}_{t,ij}^h = \left[\sum_{j'}^{\text{basal}} w_{Y,j'i}^h e_{t,j'} \right] \cdot \text{L23}_{t,ij}^h. \quad (12)$$

Finally, similarly to layer 5, we postulate layer 6 corticothalamic (CT) neurons summing up the activity of L6IT pyramidal neurons,

$$\text{L6CT}_{t,j}^h = \sum_i \text{L6IT}_{t,ij}^h, \quad (13)$$

where we note $\text{L6CT}_{t,i}^h = e_{t,i}^{q,h}$ for clarity of the neuronal mapping. This sum from L6IT to layer 6 CT neurons is spatially laid out perpendicularly to the sum of L5IT to L5ET neurons previously described.

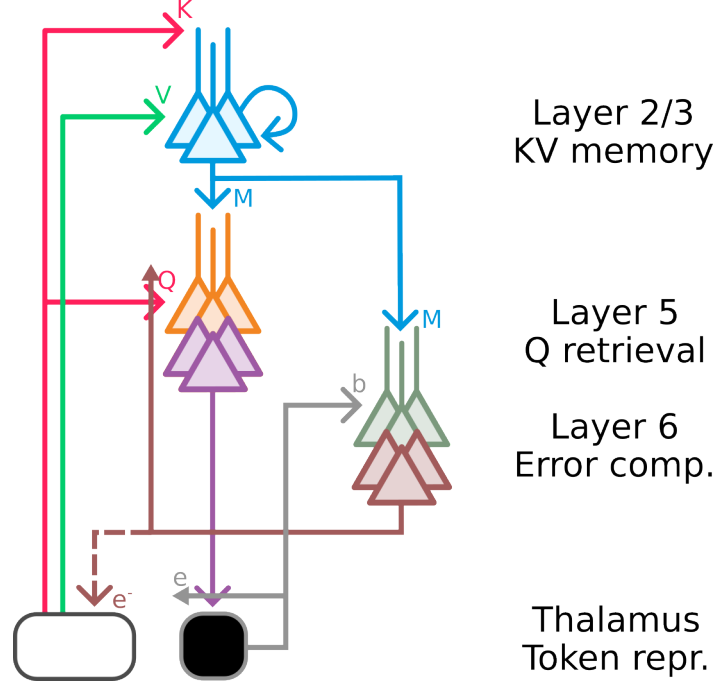


Figure 6: Error computation in cortical layer 6 for gradient-based plasticity of query weights. The current token encoded in a lower-order thalamic nucleus (white rectangle) is both integrated in a recurrent short-term associative memory encoded in layer 2/3 pyramidal cells (blue triangles) and queries or retrieves this memory in layer 5 pyramidal cells (orange and purple triangles). Both these computation are supported by core (green) and matrix (red) thalamo-cortical projections, as well as local projections from layer 2/3 to layer 5. Retrieved memory (from multiple heads, not shown) are integrated in a higher-order thalamic nucleus (black square), forming the output. This is supported by cortico-thalamic projections (purple) from layer 5 (ET, purple) pyramidal cells. This was already represented back in fig. 1. We now postulate that the output $\mathbf{y}_t$ is compared in the higher-order thalamic nucleus (black square) to a target $\mathbf{x}_{t+1}$, forming the error $\mathbf{e}_t = \mathbf{x}_{t+1} - \mathbf{y}_t$. According to eq. 10, weights $\mathbf{W}_Y^h$ of cortico-thalamic synapses sent by L5ET pyramidal cells are learned through the product of this error $\mathbf{e}_t$ and the somatic activity of the L5ET cells $\mathbf{o}_t^h$ (gray arrow with plain arrowhead). According to eq. 11, learning the weights $\mathbf{W}_Q^h$ of the thalamo-cortical synapses carrying queries requires an error first backpropagated from the thalamus to the cortex, then ‘backpropagated through’ the layer 2/3 associative memory. We have described in eqs. 12 and 13 a circuit of L6IT and CT pyramidal cells realizing this error computation. In short, thalamic errors are backpropagated to the cortex (through ‘feedback’ thalamo-cortical projections) and integrated in the basal tree of L6IT cells (teal triangles). The somatic activity of L6IT cells reflect the componentwise product of these backpropagated errors at the basal dendrites times the associative memory received at the apical dendrites from local layer 2/3 projections, see eq. 12. These are then summed up in layer 6 CT pyramidal neurons (brown triangles) to realize the matrix multiplication proper, computing $\mathbf{e}_t^{q,h}$ in the somatic activity, eq. 13. This error is then used to learn the weights of synapses computing queries (brown arrow with plain arrowhead). Additionally, these errors can be backpropagated to the lower-order thalamic nucleus to construct the lower-level thalamic error when explicit targets are not available (brown dashed arrow).

The layer 6 CT neurons have a double role. First, they modulate the strengths of synapses on the basal dendrites of L5IT neurons,

$$\dot{\mathbf{W}}_Q^h \propto \text{L6CT}_t^h \mathbf{x}_t^T, \quad (14)$$

simply rewriting eq. 11. This assumes collaterals from layer 6 CT to L5IT pyramidal cells, and indeed anatomical studies clearly show that local axonal projections of deep layer 6 pyramidal cells (where layer 6 CT neurons are enriched compared to IT neurons) end in layer 5a (see e.g. Sievers et al. 2024 for a modern account). Second, layer 6 CT neurons backproject their cortical error to the previous thalamic nucleus, forming the instantaneous component of error propagation between attention layers, $\sum_h \mathbf{W}_Q^{hT} \text{L6CT}_t^h$ (see eqs. 49 and 50). This is in

accordance with the tendency of layer 6 CT neurons to project to the same thalamic nuclei that project to their cortical area, forming thalamo-cortical loops (Cassidy, Macias, Lagos, Ugorji, and Callaway 2025). Again, this double role of the memory synapses in inference and plasticity (eqs. 5 and 12) makes the cortical implementation unique and more flexible than Hopfield-type implementations.

Reward-based gating of short cortical and long subcortical memories

We can yet learn another lesson from biology with regard to combining different forms of memory. Instead of adding memories in parallel via cross-attention at the level of apical L5 inputs (eqs 8 and 9), memories can also be stacked in series at the level of the L2/3 input to provide yet longer memory time scales. In fact, the cortical layer 2/3 working memory may be nurtured by a hippocampal episodic memory, operating itself on a longer time scale, and in combination with the recurrent L2/3 memory, bootstrapping the cortical memories (Lee, Kim, Park, and Cho 2023). Instead of calculating the value $\mathbf{v}_t^h = \mathbf{W}_V^h \mathbf{x}_t$ for the cortical area h out of the input token $\mathbf{x}_t$, the value may be delivered from the hippocampal Hopfield-type memory (Gershman, Fiete, and Irie 2025). For this, the hippocampal memory is itself queried by $\mathbf{q}_t^{\text{hc}} = \mathbf{W}_V^{\text{hc}} \mathbf{x}_t$, so that it returns the output

$$\mathbf{o}_t^{\text{hc}} = \mathbf{M}_t^{\text{hc}} \mathbf{q}_t^{\text{hc}} = \frac{1}{\tau^{\text{hc}}} \sum_{t' \leq t} \lambda_{\text{hc}}^{t-t'} \mathbf{v}_{t'}^{\text{hc}} \alpha_{t,t'}^{\text{hc}}. \quad (15)$$

as input to the basal dendrites of the cortical L23^h pyramidal neurons (Methods, eq. 24). Because the hippocampal memory time constant τ^{hc} is long, the decay factor $\lambda_{\text{hc}} = 1 - \frac{g_{\text{el}}}{\tau^{\text{hc}}}$ is close to 1. The hippocampal input $\mathbf{o}_t^{\text{hc}}$ is added to the own value generated from the thalamic input in the L2/3 basal dendrites, weighted by individual factors, $\alpha_V^h \mathbf{v}_t^h + \alpha_V^{\text{hc}} \mathbf{o}_t^{\text{hc}}$. The relative weighting depends on the discounted future reward allocated to the cortical and hippocampal memory, respectively, competitively enhanced by a softmax operation, i.e. $\alpha_V^h = \frac{e^{\hat{\mathbf{v}}_t^h}}{e^{\hat{\mathbf{v}}_t^h} + e^{\hat{\mathbf{v}}_t^{\text{hc}}}}$ and $\alpha_V^{\text{hc}} = \frac{e^{\hat{\mathbf{v}}_t^{\text{hc}}}}{e^{\hat{\mathbf{v}}_t^h} + e^{\hat{\mathbf{v}}_t^{\text{hc}}}}$, see eqs 17 and 18 below. The weighting may be implemented by dendritic input originating from basal ganglia (Takahashi, Moberg, Zolnik, Catanese, Sachdev, Larkum, and Jaeger 2021) that changes the gain of individual dendritic branches (Silver 2010).

With the cortico-hippocampal loop, the effective memory gets stacked. Instead of only writing the value-key association $\mathbf{v}_t^h (\mathbf{k}_t^h)^\top$ into the recurrent L2/3 memory (as expressed in eq. 1), the input $\mathbf{x}_t$ also loops through the hippocampal memory, and $(\alpha_V^h \mathbf{v}_t^h + \alpha_V^{\text{hc}} \mathbf{o}_t^{\text{hc}}) (\mathbf{k}_t^h)^\top$ is written into the cortical memory. When querying the L2/3 memory at time t , not only values $\mathbf{v}_{t'}^h$ at a passed time t' are retrieved, but also values $\mathbf{v}_{t''}^{\text{hc}}$ stored in hippocampus at time t'' further back and entering via $\mathbf{o}_{t'}^{\text{hc}}$, see eq. 15, in the same way as $\mathbf{v}_{t'}^h$ enters,

$$\text{L5ET}_t^h = \frac{1}{\tau^h} \sum_{t' \leq t} \lambda_h^{t-t'} \left(\alpha_V^h \mathbf{v}_{t'}^h + \alpha_V^{\text{hc}} \mathbf{o}_{t'}^{\text{hc}} \right) \alpha_{t,t'}^h. \quad (16)$$

The reward-based weighting of the memories by the factors α_V^h and α_V^{hc} , respectively, represents a fundamental departure from pure transformers, reminiscent to the introduction of reinforcement learning in large language models to incentivize reasoning (DeepSeek-AI 2025). In our case, reinforcement learning selects at each moment the memory that promises the highest reward. Supervised learning then improves the prediction of the next token based on the selected memory, making it more likely that a next time the memory is again selected by the reinforcement learning.

A corresponding gating of cortical activity by reward-mediating subcortical nuclei, as expressed in eq. 16 for the hippocampal-cortical loop, may also apply to habit and motor memories recruited via basal ganglia-cortical loop that turns through thalamic nuclei (Graybiel 1998; Yin and Knowlton 2006). In fact, neurons in the anterior lateral motor cortex maintain persistent activities that are prospective for future actions, and that are mirrored in the associated thalamic nuclei (Guo, Inagaki, Daie, Druckmann, Gerfen, and Svoboda 2017). The same thalamic nuclei receive an action motivation or ‘vigor’ signal from basal ganglia (Turner and Desmurget 2010), and the thalamo-cortical projections to the motor cortex are shown to increase the vigor of initiating planned or reward-predicting motor responses (Takahashi, Moberg, Zolnik, Catanese, Sachdev, Larkum, and Jaeger 2021).

Gain modulation in L5ET pyramidal neurons by the discounted future reward

While L2/3 memory neurons are sequentially storing all the incoming value-key pairs, the values are later on selectively read out based on the query-key alignment. The cortical output is a sum of past values, selected by the key-query alignment, $\mathbf{o}_t^h = \sum_{t' \leq t} \mathbf{v}_{t'}^h (\mathbf{k}_{t'}^h \mathbf{q}_t^h)$. The value selection is learned to best predict the next input token as a weighted sum of the cortical outputs across heads, $\mathbf{x}_{t+1} \stackrel{!}{=} \sum_h \mathbf{W}_Y^h \mathbf{o}_t^h$. In biology, however, predicting a next input is often only a subordinate goal towards the more relevant goal of predicting reward or punishment in the future. Hence, the selection of cortical output may first be narrowed down to reward-predicting outputs, and only based on this selection will the next token be predicted. Reward prediction is achieved by a value function $\mathbb{V}_t^h$ that assigns the expected discounted future reward to a given state, here the cortical output $\mathbf{o}_t^h$ of area h (Sutton and Barto 2018). We therefore postulate that the cortical readout $\mathbf{o}_t^h$ is modulated by the predicted future reward $\mathbb{V}_t^h$, and the next token is predicted based on this modulation, $\mathbf{x}_{t+1} \stackrel{!}{=} \sum_h \mathbf{W}_Y^h \mathbb{V}_t^h \mathbf{o}_t^h$.

Reward prediction and reinforcement learning in the brain is typically assigned to the basal ganglia (Doya 2000). Cortico-basal ganglia feedback loops are involved in reward-based adaptation of movements (Turner and Desmurget 2010; Leblois, Boraud, and Hansel 2025), while neurons in the motor cortex and the associated thalamic nuclei are prospective for future actions towards rewards (Guo, Inagaki, Daie, Druckmann, Gerfen, and Svoboda 2017). Consistent with these findings and the cortical-basal ganglia connectivity, we postulate that L2/3 and L5IT pyramidal neurons project to basal ganglia in order to estimate future reward, and the reward estimate is fed back to L5ET pyramidal neurons to modulate their gain (Takahashi, Ebner, Sigl-Glückner, Moberg, Nierwetberg, and Larkum 2020; Moberg and Takahashi 2022). Hence, for each cortical area (i.e. each head h) including the hippocampus, a population of basal ganglia neurons estimates the expected discounted future reward given the corresponding cortical and hippocampal activities,

$$\widehat{\mathbb{V}}_t^h \sim \mathbb{V}_t^h = \mathbb{E} \left[\sum_{t'=t}^{\infty} \gamma_h^{t'-t} R_{t'} \mid \mathbf{L23}_t^h, \mathbf{L5IT}_t^h \right] \quad (17)$$

$$\widehat{\mathbb{V}}_t^{\text{hc}} \sim \mathbb{V}_t^{\text{hc}} = \mathbb{E} \left[\sum_{t'=t}^{\infty} \gamma_{\text{hc}}^{t'-t} R_{t'} \mid \mathbf{o}_t^{\text{hc}} \right], \quad (18)$$

where $R_{t'}$ takes on values between -1 and 1 , $\gamma_{h/\text{hc}}$ is the discount factor, and $\widehat{\mathbb{V}}_t^{h/\text{hc}}$ is the basal ganglia estimate of $\mathbb{V}_t^{h/\text{hc}}$. The reward R_t is sparsely given after a sequence of produced output tokens, judging their overall performance. For instance, during a language production task, a reward may be externally applied, or internally elicited, at the end of a speech, following standard language evaluations (Papineni, Roukos, Ward, and Zhu 2002). Conditioning on the L23 memory neurons, $\mathbf{L23}_t^h \equiv \mathbf{M}_t^h$, provides memory specific information such as memory lengths, while conditioning on the readout neurons $\mathbf{L5IT}_t^h$ provides additional query-based information. Conditioning on the hippocampal output provides the expected discounted future reward of the hippocampal memory upon the given query.

The basal ganglia neurons project back (via thalamus, not modeled) to the same cortical area and modulate the gain of L5ET neurons, with the $\widehat{\mathbb{V}}_t^h$ synapses targeting the apical dendrites, and the $\mathbf{L5IT}_{t,ij}^h$ synapses (all with weights 1) targeting the somatic region,

$$\mathbf{L5ET}_{t,i}^h = \frac{\mathbf{e}^{\widehat{\mathbb{V}}_t^h}}{\sum_{h'} \mathbf{e}^{\widehat{\mathbb{V}}_t^{h'}}} \sum_j \mathbf{L5IT}_{t,ij}^h. \quad (19)$$

The normalization is assumed to be calculated at the level of the basal ganglia, e.g. by divisive inhibition of a (Wilson, Runyan, Wang, and Sur 2012; Brown, Pan, and Dudman 2014; Goodroe, Starnes, and Brown 2018). When including the hippocampal memory, $\sum_j \mathbf{L5IT}_{t,ij}^h$ in eq. 19 is replaced by the expression in eq. 16. In fact, the different values $\widehat{\mathbb{V}}_t^h$ and $\widehat{\mathbb{V}}_t^{\text{hc}}$ mainly originate in the different memory lengths associated to the various cortical areas h and the hippocampus. Prospective reward or punishment signals have in fact been observed in apical dendrites of L5 pyramidal neurons *in vivo* (Schoenfeld et al. 2024), while a multiplicative modulation of cortical (L2/3) pyramidal neurons by dopamine, the internal reward signal, was observed *in vitro* (Thurley, Senn, and Lüscher 2008).

A value function, e.g. $\mathbb{V}_t^h$, was shown to be learnable by a 2-compartment neuron based on the dendritic prediction of somatic activity (Urbanczik and Senn 2014; Brea, Gaál, Urbanczik, and Senn 2016). The dendritic compartment of the basal ganglia neurons, e.g. corresponding to area h , is driven by the L5IT^h inputs, followed by a dendritic nonlinearity,

$$\widehat{\mathbb{V}}_t^h = \phi \left(\sum_{i,j} \mathbf{W}_{\mathbb{V},ij}^{h,\text{L23}} \text{L23}_{t,ij}^h + \mathbf{W}_{\mathbb{V},ij}^{h,\text{L5}} \text{L5IT}_{t,ij}^h \right). \quad (20)$$

This dendritic voltage propagates to the soma, arriving in the attenuated form $\gamma_h \widehat{\mathbb{V}}_t^h$. During the learning process, the soma additionally is additionally nudged by the reward signal R_t , so that the overall somatic activity becomes $U_t^h = R_t + \gamma_h \widehat{\mathbb{V}}_t^h$. The cortico-basal ganglia synapses $\mathbf{W}_{\mathbb{V}}^h$ are learned by minimizing the dendritic prediction of the somatic activity, i.e. by gradient descent on the squared dendritic prediction error $\frac{1}{2}(U_t^h - \widehat{\mathbb{V}}_{t-1}^h)^2$. For instance, the synapses from the L2/3 pyramidal neurons to the basal ganglia neuron adapts according to

$$\dot{\mathbf{W}}_{\mathbb{V}}^{h,\text{L23}} \propto -\frac{d}{d\mathbf{W}_{\mathbb{V}}^{h,\text{L23}}} \frac{1}{2}(U_t^h - \widehat{\mathbb{V}}_{t-1}^h)^2 = (R_t + \gamma_h \widehat{\mathbb{V}}_t^h - \widehat{\mathbb{V}}_{t-1}^h) \phi' \text{L23}_{t-1}^h. \quad (21)$$

The parenthesis represents the temporal difference (TD) error, $\delta_{t-1}^h = R_t + \gamma_h \widehat{\mathbb{V}}_t^h - \widehat{\mathbb{V}}_{t-1}^h$, and the learning rule is just the classical TD-delta rule for estimating the expected discounted future reward (Sutton and Barto 2018). For simplicity, we assume that the basal ganglia output is the somatic voltage scaled by the inverse attenuation factor $\frac{1}{\gamma_h}$, so that – in the absence of a reward signal – the basal ganglia output is in fact $\widehat{\mathbb{V}}_t^h$. The basal ganglia neurons modulate the L5IT^h neurons as in Eq. 19.

Learning language production and translation with cortical MHSA

Following our formalization of corticothalamic MHSA circuits, with extensions to cortico-cortical cross attentions, we asked whether these circuits with the biological form of synaptic plasticity are in fact able to generate cognitive behaviours.

In a first test we learned a small network on Shakespeare’s (tiny) language data set to produce Shakespeare type language¹ (fig. 5).

In a second test we considered language translation using cortical cross-attention (see eq. 8, fig. 4 b). We learned the English-German translation of image caption using the Multi30K data set², with 4-layer MHSA networks forming encoder and decoder, linked with cross-attention (Vaswani et al. 2017). Encoder and decoder are run according to algorithm 4, but the encoder queried the key-value memory from the encoder. The input sequence $\mathbf{x}_t$ to encoder was formed by the English input tokens, while the input sequence $\mathbf{z}_t$ to the decoder consisted in the translated German tokens. These German tokens were chosen with probabilities generated by the decoder output. Note that, despite the fixed random memory matrices, learning the query and output matrices in the decoder still shapes its internal activity. For more simulation details see Methods (in preparation).

Matching to human intracortical activity during language production

We extracted a macro-connectome of the human left parietal cortex (using The Virtual Brain, TVB) and decomposed the connectome into encoder-decoder building blocks following eq. 9, see Methods. In short, the nearly all-to-all cortical macro-connectome is decomposed into a sensory and motor pathway that form the encoder and decoder. Within these pathways, forward projections are identified by the readout matrix $\mathbf{W}_Y^{XZ}$, while backward projections provide the backpropagated error via transpose, $\mathbf{e}_t^{o,XZ} = \mathbf{W}_Y^{XZ,\top} \mathbf{e}_t^z$. So far, we considered $\mathbf{W}_Y^{XZ}$ as readout matrix from area Z (modulated by the memory from area X) to a thalamic nucleus. Yet, the feedback via thalamus does not exclude direct cortico-cortical connections (e.g. via layer 4 in the next area) that we postulate here to be identical with the thalamo-cortical projections. In fact, thalamic nuclei representing ‘provincial’ hubs connect cortical areas that are themselves connected (Hwang, Bertolero, Liu, and D’Esposito 2017; Shepherd and Yamawaki 2021; Kumar, Beckmann, Scheffler, and Grodd 2022; Sherman and Usrey 2024b). Eq. 9 describes

¹tinyshakespeare.txt (Karpathy)

²<https://github.com/multi30k/dataset>

such a projection for the decoder, but it is equally present for the encoder (with cross-attention, say from the decoder, or without).

Cortical connections between the two pathways are identified by the matrix $\mathbf{A}^{XZ}$ that provide the L2/3 memory from area X from the encoder to area Z from the decoder. The error that modulates $\mathbf{W}_Q^{XY}$ is calculated as $\mathbf{e}_t^{q,XZ} = \mathbf{A}^{XZ} \mathbf{M}_t^{XZ,T} \mathbf{e}_t^{o,XZ}$. It is represented within area Z and again requires projections from the L2/3 memory neurons in area X to area Z . It is reasonable to assume that, symmetrically, the sensory pathway (an area X) is itself modulated by the motor pathway (the area Z) via cross-attention, but we leave this out. Overall, the (non-directional) macro-connectome between the TVB nodes forming the sensory and motor pathways are identified by the within-pathway connections, e.g. $\mathbf{W}_Y^{XZ}$, and the across-pathway connections, e.g. $\mathbf{A}^{XZ}$.

Discussion

Summary. We highlighted analogies between the structure of cortico-thalamic circuits and multihead, unnormalized, linear self-attention. In particular, we proposed that the pyramidal cells of layer 2/3 implement an associative key-value memory (Gershman, Fiete, and Irie 2025), while the pyramidal cells of layer 5 represent the memory retrieved by the current query. The fine-grained structure of core and matrix projections from the thalamus to the cortex, cortical gain modulation, and cell-type-specific cortical circuits are well suited to implement the underlying computation. Different cortical areas can be interpreted as different heads of attention in a transformer block that are integrated e.g. in a higher thalamic nucleus. Cross- (rather than self-) attention appears as a computation in associative cortices driven by different modalities, or as lateral or top-down projections between cortical areas that modulate the gain of pyramidal neurons in the target area. The classical encoder-decoder architecture is interpreted as a memory-modulated sensory-cognitive-motor mapping. We proposed extensions to the transformer framework to include hippocampal-cortical and basal ganglia-cortical interactions, with the computational goals of remote memory retrieval and reward-based memory enhancement. Last but not least, we showed how gradient-based learning could be neurally implemented with dedicated error neurons.

From multihead self-attention to cortico-thalamic transformer blocks. Reproducing the full computation of a transformer block entails the sequential computation of multihead self-attention and a fully-connected network with one hidden layer, together with normalization. In this work, we have covered the central innovation of transformer networks, multihead self- and cross-attention. The remaining normalization is well-studied in neuroscience and has been described as a canonical neural computation in terms of divisive normalization (Carandini and Heeger 2012; Shen, Wang, and Navlakha 2021) or soft winner-take-all over dendritic activities (Murayama, Pérez-Garci, Nevian, Bock, Senn, and Larkum 2009), but it is left out here.

Parallel and serial cross-attention from hippocampal memory. We have proposed that layer 2/3 pyramidal cells encode a short-term ‘working’ memory of the immediate history by integrating key-value pairings. Long-term memory might also rely on a key-value memory system, likely involving the hippocampal and parahippocampal structures (Whittington, Warren, and Behrens 2022; Gershman, Fiete, and Irie 2025) and their projection to cortical layer 1 (Doron et al. 2020). These hippocampal memory projections could target tuft dendrites of layer 5 pyramidal cells, leading to a parallel summation of attention signals. They may also target layer 2/3 pyramidal cells (Ledderose et al. 2023), leading to a serial stacking of attention signals.

Multimodal and cortico-cortical attention signals. We showed that thalamo-cortical projections from thalamic nuclei encoding different sensory modalities to a common associative cortical area, and interareal cortico-cortical projections, may realize a form of cross-attention. More generally, the structure of cortico-cortical feedforward and feedback connections resembles the structure of core and matrix projections in their laminar targets (Markov et al. 2014; Harris et al. 2019; Shepherd and Yamawaki 2021). The computation realized by feedforward and feedback projections might then be analyzed along the same line as thalamo-cortical projections, as participating respectively in the computation of values, and keys and queries. In support, recent evidence from direct inactivation of cortico-cortical feedback projections demonstrates their role in attentional gain modulation (Debes and Dragoi 2023).

Reward-modulated attention signals and memory enhancement. The formalization of biological observations in the transformer analogy inspires computational architectures that so far have not been considered in the technical applications. The synaptic real-time learning and the hippocampal memory stacking are such

examples. Another salient example is the reward-predictive gain modulation of cortical pyramidal neurons that jointly acts with the attentional gain modulation on the cortical representations (Schoenfeld et al. 2024). In the biology, the reward-modulation further enhances task-relevant representations that become behaviorally relevant at later stages (Guo, Inagaki, Daie, Druckmann, Gerfen, and Svoboda 2017). How much these biological features boost the computational efficacy of artificial applications remains to be investigated.

Over-expressed attention signals and brain diseases. Given the human-like performance of transformers in solving some cognitive and behavioral tasks, together with the suggested embedding into cortical and subcortical structures, we may further ask how cognitive or behavioral abnormalities are caused by cortical or subcortical abnormalities. Since it is the various attention signals that equip transformers with their cognitive power, the trace to explain cognitive dysfunctions hints to deviant forms of attention. For instance, epilepsy may be explained by a too strong layer 2/3 recurrency (e.g. via over-expressed NMDA receptors), schizophrenia by a too strong self-attention on past memories instead of on current sensory inputs, or Parkinson disease by a deviant vigor (‘motor attention’) originating in reduced reward (dopamine) modulation in the basal ganglia feedback loop (citations). Inappropriate error representations in the postulated cortical layer 6 error neurons may impair learning and error-correction capabilities that, too, have yet to be linked to experimental or clinical data.

Methods

Linear self-attention

Let $\mathbf{x}_1, \dots, \mathbf{x}_T$ be a sequence of T elements or tokens, with $\mathbf{x}_t \in \mathbb{R}^d$. The self-attention mechanism central in transformer networks relies on three linear projections of sequence elements

$$\mathbf{q}_t = \mathbf{W}_Q \mathbf{x}_t \ ; \ \mathbf{k}_t = \mathbf{W}_K \mathbf{x}_t \ ; \ \mathbf{v}_t = \mathbf{W}_V \mathbf{x}_t \ , \quad (22)$$

with $\mathbf{q}_t \in \mathbb{R}^{d_k}$, $\mathbf{k}_t \in \mathbb{R}^{d_k}$, $\mathbf{v}_t \in \mathbb{R}^{d_v}$, $\mathbf{W}_Q \in \mathbb{R}^{d_k \times d}$, $\mathbf{W}_K \in \mathbb{R}^{d_k \times d}$, and $\mathbf{W}_V \in \mathbb{R}^{d_v \times d}$.

We consider computation with causal masking, that is, where a token at time t can only attend to other tokens at times $t' \leq t$ in the past. A classical self-attention mechanism (e.g. Vaswani et al. 2017) uses the softmax, or normalized exponential dot-product, as a similarity measure. It outputs the weighted sum across the T past time steps,

$$\mathbf{o}_t = \sum_{t' \leq t} \mathbf{v}_{t'} \frac{\mathbf{e}^{\mathbf{k}_{t'}^\top \mathbf{q}_t}}{\sum_{t'' \leq t} \mathbf{e}^{\mathbf{k}_{t''}^\top \mathbf{q}_t}} . \quad (23)$$

In this work, we rather use the alternative *linear* self-attention formulation, replacing the exponential nucleus with a linear product of feature maps $\phi : \mathbb{R}^{d_k} \rightarrow \mathbb{R}^{d_f}$ (Katharopoulos, Vyas, Pappas, and Fleuret 2020; Schlag, Irie, and Schmidhuber 2021). Multiple works have shown how to approximate eq. 23 arbitrarily closely using projections to random features (Peng, Pappas, Yogatama, Schwartz, Smith, and Kong 2021; Choromanski et al. 2021). Following recent work on linear self-attention, we additionally drop the normalization term (Schlag, Irie, and Schmidhuber 2021; Qin, Han, Sun, Li, Kong, Barnes, and Zhong 2022; Fu, Dao, Saab, Thomas, Rudra, and Ré 2023; Von Oswald, Niklasson, Randazzo, Sacramento, Mordvintsev, Zhmoginov, and Vladymyrov 2023; Sun et al. 2023; Yang, Wang, Zhang, Shen, and Kim 2024). We also introduce a memory discount factor $\lambda = 1 - \frac{g_\ell}{\tau} \in [0, 1]$ emerging from the leak g_ℓ and the effective integration time constant τ , corresponding to the memory length $T \sim \frac{1}{\lambda}$ in eq. 23. The outputs are then a sum of discounted and (self-)attention-weighted values,

$$\mathbf{o}_t = \frac{1}{\tau} \sum_{t' \leq t} \lambda^{t-t'} \mathbf{v}_{t'} \phi(\mathbf{k}_{t'})^\top \phi(\mathbf{q}_t) = \frac{1}{\tau} \sum_{t' \leq t} \lambda^{t-t'} \mathbf{v}_{t'} \alpha_{t',t} \ , \quad (24)$$

with attention $\alpha_{t',t} = \phi(\mathbf{k}_{t'})^\top \phi(\mathbf{q}_t)$ devoted at time t to the memory item from time t' . One advantage of non-normalized output (eq. 24) over the softmax-normalized output (eq. 23) is that, without normalization, the terms can be split into the product of a memory times the query,

$$\mathbf{o}_t = \frac{1}{\tau} \left[\sum_{t' \leq t} \lambda^{t-t'} \mathbf{v}_{t'} \phi(\mathbf{k}_{t'})^\top \right] \phi(\mathbf{q}_t) = \mathbf{M}_t \phi(\mathbf{q}_t) . \quad (25)$$

In the last equation, we introduced the key-value memory $\mathbf{M}_t = \frac{1}{\tau} \sum_{t' \leq t} \lambda^{t-t'} \mathbf{v}_{t'} \phi(\mathbf{k}_{t'})^\top$, that can also be recurrently defined as in eq. 1. An intuition for the memory parametrization can be given as a discretization of the continuous dynamics $\tau \frac{d}{dt} \mathbf{M}_t = -g_\ell \mathbf{M}_t + \mathbf{v}_t \mathbf{k}_t^\top$.

Our eq. 1 can be seen as gradient-descent learning of the memory loss $\mathcal{L}_t(\mathbf{M}) = \frac{1}{2} \|\mathbf{v}_t \phi(\mathbf{k}_t)^\top - \mathbf{M}\|^2$. A similar delta rule, but on the level of the values $\mathbf{v}_t$ instead of the memory $\mathbf{M}_t$, has been proposed by Schlag, Irie, and Schmidhuber (2021), $\mathbf{M}_t = \mathbf{M}_{t-1} + \lambda_t (\mathbf{v}_t - \mathbf{M}_{t-1} \phi(\mathbf{k}_t)) \phi(\mathbf{k}_t)^\top$. A simpler version of the memory update was suggested by retentive networks (Sun et al. 2023), $\mathbf{M}_t = \lambda \mathbf{M}_{t-1} + \mathbf{v}_t \phi(\mathbf{k}_t)^\top$, that is identical to ours for $\tau = 1$ and $\lambda = 1 - g_\ell$.

The classical formulation of eq. 23 bears a computational cost that scales quadratically in time (sequence length), since each new query is compared with all the previous keys. In contrast, the rearrangement in eq. 25 only scales linearly in time, since each key-value pairing (outer product) can be computed only once, stored in a key-value memory, and reused for every query (Katharopoulos, Vyas, Pappas, and Fleuret 2020).

Hierarchy of content-lock-probe (VKQ) memories Biological memories are distributed in space, e.g. by a hippocampal episodic memory and a cortical declarative memory, allowing to bootstrap the memories. This is achieved by sequentially stacking them, so that during the writing into the first memory the second memory is recruited, and the new item together with the second memory are jointly written into the first memory (see eq. 16). We may bootstrap a single memory with itself, formally by replacing the value with the retrieved output of the memory,

$$\mathbf{o}_t = \frac{1}{\tau} \left[\sum_{t' \leq t} \lambda^{t-t'} (\mathbf{v}_{t'} + \mathbf{o}_{t'}) \phi(\mathbf{k}_{t'})^\top \right] \phi(\mathbf{q}_t), \quad (26)$$

where in eq. 25 we added to $\mathbf{v}_{t'}$ the expression $\mathbf{o}_{t'}$ from eq. 24, evaluated at past time t' so that values $\mathbf{v}_{t''}$ enter in eq. 26 for times $t'' \leq t' \leq t$. Alternatively, individual components of the sum across t' in eq. 24 can be seen as Hebbian plasticity of the $\mathbf{W}_Q$ synapses (as suggested for the hippocampal CA3 synapses, see e.g. Gershman, Fiete, and Irie 2025), showing that the value-key-query principle can also be applied at the receptor level – here typically interpreted as a content-lock-probe principle. It is tempting to further replace $\mathbf{v}_{t''}$ by $\mathbf{o}_{t''}$, with reference to the genetic cascade triggered by the postsynaptic receptor activation (e.g. with $\mathbf{v}_{t''}$ as mRNA receptor coding sequence – the content, $\mathbf{k}_{t''}$ as promotor sequence CRE – the lock, and $\mathbf{q}_{t''}$ as activated transcription factor pCREB – the probe). The memory mixing ($\mathbf{v}_{t'} + \mathbf{o}_{t'}$) in eq. 26 may be further generalized to a weighted mix $(\alpha_V \mathbf{v}_{t'} + (1-\alpha_V) \mathbf{o}_{t'})$ in order to select the memory that promises most reward (see below).

Multihead linear self-attention

The preceding subsection describes the computation in a single head of linear self-attention. We now extend to n_h heads. The classical presentation of multihead self-attention proposes to concatenate the outputs $\mathbf{o}_t^h$ of the different heads and multiplies the resulting vector by output weight $\mathbf{W}_Y \in \mathbb{R}^{d \times n_h d_v}$, yielding the output

$$\mathbf{y}_t = \mathbf{W}_Y [\mathbf{o}_t^1 | \dots | \mathbf{o}_t^{n_h}]^\top = \sum_h \mathbf{W}_Y^h \mathbf{o}_t^h. \quad (27)$$

The last equation highlights attention heads as independent and additively integrated with head-specific output weights $\mathbf{W}_Y^h \in \mathbb{R}^{d \times d_v}$.

Overall, the output $\mathbf{y}_t$ in eq. 27 can be written as a sum of gain-modulated L5IT pyramidal neurons, summed up in L5ET output neurons in rows, and weighted by the output synapses,

$$y_{t,i} = \sum_h \sum_j W_{Y,ij}^h o_{t,j}^h = \sum_h \sum_j W_{Y,ij}^h \left[\overbrace{\sum_k M_{t,jk}^h \phi(q_{t,k}^h)}^{\text{L5ET}} \right]. \quad (28)$$

Finally, rather than having a single time constant τ , we opt to make time constants τ^h head-specific, to allow each head to integrate information at different temporal rates (and to be selected based on the expected discounted future reward for each head/area, see eq. 19).

Memory selection based on reward prediction

The step from the softmax-normalized output (eq. 23) to the non-normalized output (eq. 24) reduced the memory size from $O(T d_v d_k)$ to $O(d_v d_k)$. Yet, the memory size may be further reduced to $O(d)$ for a common dimension d of the value, key and query vectors, if these vectors are individually memorized in leaky (nonlinear) memories while only being mixed during the readout (Peng et al. 2023). Why did nature not find this memory saving solution? In fact, saving costs by reducing the memory size also reduces the memory retrieval capacity, and the brain may weigh the retrieval benefits higher than the costs.

The softmax operation applied to the attention factors $\mathbf{k}_{t'}^\top \mathbf{q}_t$ in eq. 23 is a way to sharpen the retrieval of past values $\mathbf{v}_{t'}$ from the memory. Yet, while the memory selection is based on the alignment of the current query with past keys, the brain may also select memories based on how much they contribute to future reward (Shenhav, Botvinick, and Cohen 2013). Hence, we suggest a memory selection based on the estimated discounted future reward given a memory state ($\hat{\mathbf{v}}_t^h$, see eqs 17 and 18), sharpened by the softmax function,

$$\mathbf{y}_t = \sum_h \mathbf{W}_Y^h \mathbf{o}_t^h \frac{\mathbf{e}^{\hat{\mathbf{v}}_t^h}}{\sum_{h'} \mathbf{e}^{\hat{\mathbf{v}}_t^{h'}}}. \quad (29)$$

The competitive memory selection is observed in humans depending on the task (Poldrack, Clark, Paré-Blagoev, Shohamy, Moyano, Myers, and Gluck 2001), while the expected value of the selection was suggested to be calculated in the basal ganglia (Frank, Loughry, and Reilly 2001). An analogous reward-based selection is suggested in recruiting hippocampal memories (see eqs 16 and 26).

Alternative. The work by Schoenfeld et al. 2024 and also of Bernklau, Righetti, Mehrke, and Jacob 2024 suggests to replace $\hat{\mathbf{v}}_t^h$ in eq. 29 by the TD-delta $\delta_t^h = R_t + \gamma_h \hat{\mathbf{v}}_t^h - \hat{\mathbf{v}}_{t-1}^h$ from eq. 21, leading to

$$\mathbf{y}_t = \sum_h \mathbf{W}_Y^h \mathbf{o}_t^h \frac{\mathbf{e}^{|\delta_t^h|}}{\sum_{h'} \mathbf{e}^{|\delta_t^{h'}|}}. \quad (30)$$

In fact, the gain of L5 pyramidal neurons was shown to be modulated by the absolute value (here modelled by the exponential) of the TD-error (Schoenfeld et al. 2024), while the activity of dopamine neurons (that have non-negative firing rates) are better modelled by the exponentiated TD-delta function, rather than by the TD-delta itself (Schultz, Dayan, and Montague 1997; Bernklau, Righetti, Mehrke, and Jacob 2024).

Vasanth, can you check which one performs better, eq. 29 or eq. 30? In fact, it makes much more sense to switch to the memory at the moment when that predicts more future reward than before, and do that in competition with other memories. When one would switch only right before the reward (when the value is highest), it would not make sense anymore to switch, since the crucial tokens that make the reward-difference have been selected already earlier in time, namely at the time when the TD-delta was big. Selecting the a head based on the TD-delta also prevents the decay of the gradient in time, since the ‘surprise’ signal is large at the first occurrence of a cue that predicts future reward (Schultz, Dayan, and Montague 1997).

Formal gradients for the self-attention parameters

A mechanistic theory of cognition includes the implementation of gradient-based synaptic plasticity (Richards et al. 2019; Richards and Kording 2023). As we show, the explicit memory representation by our L2/3 neurons also allows for reassembling the appropriate learning signals required to implement gradient plasticity for the query- and readout-synapses. We derive the gradients of multihead self-attention with respect to $\mathbf{W}_Y^h$, $\mathbf{W}_V^h$, $\mathbf{W}_K^h$, and $\mathbf{W}_Q^h$ of a mean squared error loss in a self-supervised, autoregressive setting. Formally, the loss is defined as a squared mismatch between the output $\mathbf{y}_t$ and the next input $\mathbf{x}_{t+1}$ that should be prospectively predicted by the self-attention network,

$$\mathcal{L}_t = \frac{1}{2} \|\mathbf{x}_{t+1} - \mathbf{y}_t\|^2 = \frac{1}{2} \|\mathbf{e}_t\|^2. \quad (31)$$

with $\mathbf{y}_t$ defined by eq. 3. Such a loss function based on prospective prediction errors was recently suggested in the form of a *least action principle* for cortical computation (Senn et al. 2024), along with a dendritic implementation of the emerging learning rules (Urbanczik and Senn 2014; Sacramento, Bengio, Costa, and Senn 2018).

The formal derivatives of $\mathcal{L}_t$ with respect to parameters are

$$-\partial_{\mathbf{W}_Y^h} \mathcal{L}_t = \mathbf{e}_t \mathbf{o}_t^{h\top}, \quad (32)$$

$$-\partial_{\mathbf{W}_V^h} \mathcal{L}_t = \frac{1}{\tau^h} \sum_{t' \leq t} \lambda_h^{t-t'} \{ \phi(\mathbf{q}_t^h)^\top \phi(\mathbf{k}_{t'}^h) \} \mathbf{b}_t^h \mathbf{x}_{t'}^\top. \quad (33)$$

$$-\partial_{\mathbf{W}_K^h} \mathcal{L}_t = \frac{1}{\tau^h} \sum_{t' \leq t} \lambda_h^{t-t'} \{ \mathbf{b}_t^{h\top} \mathbf{v}_{t'}^h \} [\phi'(\mathbf{k}_{t'}^h) \odot \phi(\mathbf{q}_t^h)] \mathbf{x}_{t'}^\top, \quad (34)$$

$$-\partial_{\mathbf{W}_Q^h} \mathcal{L}_t = \frac{1}{\tau^h} \sum_{t' \leq t} \lambda_h^{t-t'} \{ \mathbf{v}_{t'}^{h\top} \mathbf{b}_t^h \} [\phi'(\mathbf{q}_t^h) \odot \phi(\mathbf{k}_{t'}^h)] \mathbf{x}_t^\top, \quad (35)$$

with $\mathbf{b}_t^h = \mathbf{W}_Y^{h\top} \mathbf{e}_t$ the backpropagated error, and $\lambda_h^{t-t'} = (1 - \frac{g_\ell}{\tau^h})^{t-t'}$ a scalar discounting factor. We checked these results against numerical approximations based on finite differences and forward automatic differentiation³. These gradients consist of outer products, scaled by scalars and a scalar product in the case of eqs. 33 to 35. Each component of the gradient matrices consist of a product of pre- and post- synaptic quantities, respecting the notion of locality.

The temporal summations over past tokens indicate slow variables influencing synaptic plasticity, stored and incremented with the presentation of each new token. We can write eqs. 33 to 35 in a recurrent form (using that the scalar products in the curly brackets $\{\cdot\}$ yield numbers, and hence commute with the corresponding subsequent column vector). For linear ‘dendritic’ transfer functions, $\phi(x) = x$ (such as in Sun et al. 2023; Von Oswald, Niklasson, Randazzo, Sacramento, Mordvintsev, Zhmoginov, and Vladymyrov 2023, etc.), we get

$$-\partial_{\mathbf{W}_V^h} \mathcal{L}_t = \mathbf{b}_t^h \mathbf{q}_t^{h\top} \mathbf{S}_{V,t}^h, \quad (36)$$

$$-\partial_{\mathbf{W}_K^h} \mathcal{L}_t = \mathbf{q}_t^h \mathbf{b}_t^{h\top} \mathbf{S}_{K,t}^h, \quad (37)$$

$$-\partial_{\mathbf{W}_Q^h} \mathcal{L}_t = \mathbf{M}_t^{h\top} \mathbf{b}_t^h \mathbf{x}_t^\top, \quad (38)$$

with matrix-valued slow synaptic variables $\mathbf{S}_{\{V,K,Q\}}^h$ being recurrently updated according to

$$\mathbf{S}_{V,t}^h = \mathbf{S}_{V,t-1}^h + \frac{1}{\tau^h} (\mathbf{k}_t^h \mathbf{x}_t^\top - g_\ell \mathbf{S}_{V,t-1}^h), \quad (39)$$

$$\mathbf{S}_{K,t}^h = \mathbf{S}_{K,t-1}^h + \frac{1}{\tau^h} (\mathbf{v}_t^h \mathbf{x}_t^\top - g_\ell \mathbf{S}_{K,t-1}^h). \quad (40)$$

Local plasticity rules

The above energy gradients lead to the gradient-based plasticity of the corresponding synaptic weights from a presynaptic neuron j to a postsynaptic neuron i . For the readout and query-encoding synapses, the rules read as

$$\dot{W}_{Y,ij}^h \propto e_i o_j^h, \quad (41)$$

$$\dot{W}_{Q,ij}^h \propto \left[\sum_{j'} M_{j'i}^h b_{j'}^h \right] x_j. \quad (42)$$

All the quantities are instantaneous function of time (for notational convenience we skipped the index t).

Backpropagation across multiple cortico-thalamic loops

Walter’s version: Crucially, if we consider the biological form of error backpropagation (Sacramento, Bengio, Costa, and Senn 2018; Max, Kriener, Pineda García, Nowotny, Jaras, Senn, and Petrovici 2024), here from the higher thalamic nucleus to the cortical areas h , $\mathbf{b}_t^h = \mathbf{W}_Y^{h\top} \mathbf{e}_t$, then the plasticity rules for $\mathbf{W}_Y$ and $\mathbf{W}_Q$ (eqs 41 and 42) also allow for a biological implementation across multiple cortical areas and thalamic nuclei. In fact, subsequent thalamic activities are mediated by $\mathbf{x}_t = \sum_{h-} \mathbf{W}_Y^{h-} \mathbf{M}_t^{h-} \mathbf{W}_Q^{h-} \mathbf{x}_t^-$ and

³<https://github.com/arnogranier/Formal-MHSA-gradient-numeric>

$\mathbf{y}_t = \sum_h \mathbf{W}_Y^h \mathbf{M}_t^h \mathbf{W}_Q^h \mathbf{x}_t$, via cortical areas $\{\mathbf{X}^{h-}\}_{h-}$ and then $\{\mathbf{X}^h\}_h$. The memory matrix can be written as $\mathbf{M}_t^h = \mathbf{W}_V(\sum_{t' \leq t} \gamma^{t-t'} \mathbf{x}_{t'} \mathbf{x}_{t'}^\top) \mathbf{W}_K^\top$. The token $\mathbf{x}_t$ is calculated analogously in the preceding self-attention block out of the previous token, $\mathbf{x}_t = \sum_{h-} \mathbf{W}_Y^{h-} \mathbf{M}_t^{h-} \mathbf{W}_Q^{h-} \mathbf{x}_t^-$, with analogously defined matrices. We consider the loss $\mathcal{L}_t = \frac{1}{2} \|\mathbf{x}_{t+1}^- - \mathbf{y}_t\|^2 = \frac{1}{2} \|\mathbf{e}_t\|^2$, and want to calculate the gradient of $\mathcal{L}_t$ with respect to $\mathbf{W}_Y^-$. We then calculate

$$-\frac{\partial \mathcal{L}_t}{\partial \mathbf{W}_Y^{h-}} = \sum_h \frac{\partial \mathcal{L}_t}{\partial \mathbf{o}_t^{h-}} \frac{\partial \mathbf{o}_t^{h-}}{\partial \mathbf{x}_t} \frac{\partial \mathbf{x}_t}{\partial \mathbf{W}_Y^{h-}} + \sum_h \frac{\partial \mathcal{L}_t}{\partial \mathbf{o}_t^{h-}} \frac{\partial \mathbf{o}_t^{h-}}{\partial \mathbf{M}_t^{h-}} \sum_{t' \leq t} \frac{\partial \mathbf{M}_t^{h-}}{\partial \mathbf{x}_{t'}} \frac{\partial \mathbf{x}_{t'}}{\partial \mathbf{W}_Y^{h-}} \quad (43)$$

$$= \underbrace{\left(\sum_h \mathbf{W}_Q^{h\top} \mathbf{M}_t^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t \right) \mathbf{o}_t^{h-\top}}_{\text{Query path}} + \underbrace{(\Psi_t + \Psi_t^\top) \sum_{t' \leq t} \gamma^{t-t'} \mathbf{x}_{t'} \mathbf{o}_{t'}^{h-\top}}_{\text{Memory path}} \quad (44)$$

with outputs $\mathbf{o}_t^{h-} = \mathbf{M}_t^{h-} \mathbf{W}_Q^{h-} \mathbf{x}_t^-$ in the upstream block and ‘memory error’ $\Psi_t = \sum_h \epsilon_t^h \tilde{\mathbf{q}}_t^{h\top}$, where $\epsilon_t^h = \mathbf{W}_V^{h\top} \mathbf{b}_t^h = \mathbf{W}_V^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t$ and $\tilde{\mathbf{q}}_t^h = \mathbf{W}_K^\top \mathbf{W}_Q^h \mathbf{x}_t$ is the error and activity information, respectively, backpropagated to the memory. We can also write $\Psi_t = \sum_h \mathbf{W}_V^{h\top} \mathbf{b}_t^h \mathbf{q}_t^{h\top} \mathbf{W}_K^h$. Remember that $\mathbf{x}_{t'} = \sum_{h-} \mathbf{W}_Y^{h-} \mathbf{o}_{t'}^{h-}$.

To iterate the backpropagation to yet the next upstream self-attention block, we define the instantaneous error

$$\mathbf{e}_t^- = \underbrace{\sum_h \mathbf{W}_Q^{h\top} \mathbf{M}_t^{h\top} \mathbf{b}_t^h}_{\text{Query error}} + \underbrace{(\Psi_t + \Psi_t^\top) \mathbf{x}_t}_{\text{Memory error}}. \quad (45)$$

Out of this error $\mathbf{e}_t^-$ the upstream block calculates the plasticity in the same way as given by gradient formula in eq. 44, but with $\mathbf{e}_t$ replaced by $\mathbf{e}_t^-$ and with the weights inheriting an upper index $-$. Notice that the query term contains the query error $\mathbf{M}_t^{h\top} \mathbf{b}_t^h$ also entering as error in the plasticity rule for $\mathbf{W}_Q$, see 38. The memory term is composed of the error term $(\Psi_t + \Psi_t^\top)$ times the synaptic eligibility trace that is itself the low-pass filtered product of pre- and postsynaptic activities.

The plasticity in that preceding thalamo-cortical forward loop, $\mathbf{x}^- \rightarrow \{\mathbf{X}^{h-}\}_{h-} \rightarrow \mathbf{x}$, is then analogous to the one in eqs 41 and 42, but with quantities indexed by $^{h-}$ or $^-$, like $\mathbf{b}_t^{h-} = \mathbf{W}_Y^{h-, \top} \mathbf{e}_t^-$. The weights $\mathbf{W}_V^{h-}$ and $\mathbf{W}_K^{h-}$ may again be fixed to random values.

Arno’s version: In this section we first complete our derivation of formal gradients for a multihead self-attention layer by providing the gradient of a squared error loss (eq. 31) with respect to current layer input $\mathbf{x}_t$. This gives the form of instantaneous error propagation from one layer to the previous,

$$-\partial_{\mathbf{x}_t} \mathcal{L}_t = \sum_h \mathbf{W}_Q^{h\top} \mathbf{e}_t^{q,h} + \sum_h \frac{1}{\tau^h} \left[\{\mathbf{k}_t^{h\top} \mathbf{q}_t^h\} \mathbf{W}_V^{h\top} \mathbf{b}_t^h + \{\mathbf{v}_t^{h\top} \mathbf{b}_t^h\} \mathbf{W}_K^{h\top} \mathbf{q}_t^h \right], \quad (46)$$

with again $\mathbf{e}_t^{q,h} = \mathbf{M}_t^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t$ (the error term in the plasticity rule of $\mathbf{W}_Q^h$, see eq. 11). The first term of the right hand side involves propagation through the short-term memory ($\mathbf{M}_t^h$, in the definition of $\mathbf{e}_t^{q,h}$), then through the query weights ($\mathbf{W}_Q^h$). The second term involves backpropagation through value and key weights ($\mathbf{W}_V^h$ and $\mathbf{W}_K^h$). The crucial step in the derivation of eq. 46 is to write $\mathbf{M}_t^h$ as a function of $\mathbf{M}_{t-1}^h$ and $\mathbf{x}_t$ following eq. 1, then to remark that $\mathbf{M}_{t-1}^h$ is independent of $\mathbf{x}_t$. The validity of the formal derivation was again checked against a numerical approximation¹. The plasticity rules for $\mathbf{W}_{\{K,Q,V\}}$ of a layer below are analogous to eqs. 33 to 35 but with $-\partial_{\mathbf{x}_t} \mathcal{L}_t$ in place of $\mathbf{e}_t = -\partial_{\mathbf{y}_t} \mathcal{L}_t$ (not checked).

A symmetry in the second term of the right hand side can be highlighted by rewriting

$$-\partial_{\mathbf{x}_t} \mathcal{L}_t = \boldsymbol{\xi}_t + (\Psi_t + \Psi_t^\top) \mathbf{x}_t, \quad (47)$$

with $\boldsymbol{\xi}_t = \sum_h \mathbf{W}_Q^{h\top} \mathbf{e}_t^{q,h}$ and $\Psi_t = \sum_h \frac{1}{\tau^h} \mathbf{W}_V^{h\top} \mathbf{b}_t^h \mathbf{q}_t^{h\top} \mathbf{W}_K^h$.

Finally, to complete the picture, we consider the case of propagation through two layers and derive the formal gradient of the bottom output weights. We introduce the layer index ℓ , such that $\mathbf{x}_\ell = \sum_h \mathbf{W}_{Y,\ell}^h \mathbf{M}_{t,\ell}^h \mathbf{W}_{Q,\ell}^h \mathbf{x}_{t,\ell-1}$

and rewrite the loss for two-layers propagation $\mathcal{L}_t = \|\mathbf{x}_{t+1,\ell-2} - \mathbf{x}_{t,\ell}\|^2$. The gradient with respect to the bottom output weights $\mathbf{W}_{Y,\ell-1}^h$ is

$$-\partial_{\mathbf{W}_{Y,\ell-1}^h} \mathcal{L}_t = \sum_h \mathbf{W}_{Q,\ell}^{h\top} \mathbf{e}_{t,\ell}^{q,h} \mathbf{o}_{t,\ell-1}^{h\top} + \sum_h \frac{1}{\tau^h} \sum_{t' \leq t} \lambda_h^{t-t'} \left[\{\mathbf{k}_{t',\ell}^{h\top} \mathbf{q}_{t,\ell}^h\} \mathbf{W}_{V,\ell}^{h\top} \mathbf{b}_{t,\ell}^h + \{\mathbf{v}_{t',\ell}^{h\top} \mathbf{b}_{t,\ell}^h\} \mathbf{W}_{K,\ell}^{h\top} \mathbf{q}_{t,\ell}^h \right] \mathbf{o}_{t',\ell-1}^{h\top}, \quad (48)$$

where again $\mathbf{o}_{t,\ell-1} = \mathbf{M}_{t,\ell-1}^h \mathbf{W}_{Q,\ell-1}^h \mathbf{x}_{t,\ell-2}$ and $\lambda_h^{t-t'} = (1 - \frac{g_\ell}{\tau^h})^{t-t'}$. Again, this was checked numerically against automatic differentiation¹. Equation 48 has the same structure as eq. 46, with an additional decaying sum over past times in the second term of the right hand side and outer products with head outputs $\mathbf{o}_{t',\ell-1}^h$. In fact, in the second term we find the same structure of outer products scaled by scalar products as in eqs. 33 to 35, that can again be interpreted as scaled products of presynaptic activity ($\mathbf{o}_{t',\ell-1}^h$) times post-synaptic ‘error’. Using the same abstractions as eq. 47, we can rewrite

$$-\partial_{\mathbf{W}_{Y,\ell-1}^h} \mathcal{L}_t = \boldsymbol{\xi}_{t,\ell} \mathbf{o}_{t,\ell-1}^{h\top} + (\boldsymbol{\Psi}_{t,\ell} + \boldsymbol{\Psi}_{t,\ell}^\top) \mathbf{S}_{Y,t,\ell}^h, \quad (49)$$

with the eligibility trace $\mathbf{S}_{Y,t,\ell}^h = \sum_{t' \leq t} \lambda_h^{t-t'} \mathbf{x}_{t',\ell} \mathbf{o}_{t',\ell-1}^{h\top}$ that can again be written as a leaky integration of the product of pre- and post-synaptic (somatic) activity,

$$\mathbf{S}_{Y,t,\ell}^h = (1 - \frac{g_\ell}{\tau^h}) \mathbf{S}_{Y,t-1,\ell}^h + \mathbf{x}_{t,\ell} \mathbf{o}_{t,\ell-1}^{h\top}. \quad (50)$$

Biologically, the first term in eq. 49 can easily be implemented by backpropagating the activity of layer 6 CT pyramidal cells to the afferent thalamic nucleus, as detailed in the corresponding section of the Results. The second term again hints at the existence of slow synaptic variables that integrate the product of pre- and post-synaptic activity on the timescale of the short-term memory.

Simulation details

Here are some hints for the simulations. Because our plasticity rules are gradient decent (eqs. 32 to 35, cross-checked⁴, we run the PyTorch built-in training with adapting learning rates.

Pseudocodes annotated with the proposed biological substrates

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⁴<https://github.com/arnogranier/Formal-MHSA-gradient-numeric>

Algorithm 1 Multihead linear self-attention in cortico-thalamic circuits — One Layer, overview

```

for each head/area  $h$  do
  Calculate keys, values, and queries: ▷ Thalamo-cortical
     $\mathbf{k}_t = \mathbf{W}_K^h \mathbf{x}_t, \mathbf{v}_t = \mathbf{W}_V^h \mathbf{x}_t, \mathbf{q}_t = \mathbf{W}_Q^h \mathbf{x}_t$ 
  Update key-value memory (eq. 1): ▷ Cortical layer 2/3
     $\mathbf{M}_t^h = \mathbf{M}_{t-1}^h + \frac{1}{\tau^h} (\mathbf{v}_t^h \phi(\mathbf{k}_t^h)^\top - g_\ell \mathbf{M}_{t-1}^h)$ 
  Retrieved memory from cortical area  $h$  (eq. 3): ▷ Cortical layer 5
     $\mathbf{o}_t^h = \mathbf{M}_t^h \phi(\mathbf{q}_t^h)$ 
end for
Output computation by summing across heads
   $\mathbf{y}_t = \sum_h \mathbf{W}_Y^h \mathbf{o}_t^h$  ▷ Cortico-thalamic

Error computation
   $\mathbf{e}_t = \mathbf{x}_{t+1} - \mathbf{y}_t$  ▷ Thalamic error

for each head/area  $h$  do
  Update output weights (eq. 10):
     $\dot{\mathbf{W}}_Y^h \propto \mathbf{e}_t \mathbf{o}_t^{h\top}$  ▷ Gradient-based synaptic plasticity
  Query error computation: ▷ Thalamo-cortical and cortical layer 6
     $\mathbf{e}_t^{q,h} = \mathbf{M}_t^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t$ 
  Update query weights (eq. 11):
     $\dot{\mathbf{W}}_Q^h \propto \mathbf{e}_t^{q,h} \mathbf{x}_t^\top$  ▷ Gradient-based synaptic plasticity
end for

```

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Algorithm 2 Multihead linear Self-Attention in cortico-thalamic circuits — One layer, Details

Require: $\mathbf{x} \in \mathbb{R}^d$ ▷ Current token — lower-order thalamus
Require: $\mathbf{t} \in \mathbb{R}^d$ ▷ Target, usually a prospective $\mathbf{x}$
Require: $\forall h, \mathbf{L23}^h \in \mathbb{R}^{d_v \times d_k}$ ▷ Previous layer 2/3 memory matrices for all heads (0 at the first iteration)
Require: $\forall h, \mathbf{W}_V^h \in \mathbb{R}^{d_v \times d}$ ▷ Thalamo-cortical value weights for all heads
Require: $\forall h, \mathbf{W}_K^h \in \mathbb{R}^{d_k \times d}$ ▷ Thalamo-cortical key weights for all heads
Require: $\forall h, \mathbf{W}_Q^h \in \mathbb{R}^{d_k \times d}$ ▷ Thalamo-cortical query weights for all heads
Require: $\forall h, \mathbf{W}_Y^h \in \mathbb{R}^{d \times d_v}$ ▷ Cortico-thalamic output weights for all heads
Require: $\forall h, \tau^h \in [1, \infty)$ ▷ Time constants for all heads
Require: $g_\ell \in [0, 1]$ ▷ Leak factor
Require: $\eta_Y \in \mathbb{R}, \eta_Q \in \mathbb{R}$ ▷ Learning rates

Inference

$\mathbf{y} \leftarrow \text{zeros}(d)$

for $h = 1, \dots, n_h$ **do**

$\mathbf{v}^h \leftarrow \mathbf{W}_V^h \mathbf{x}$ ▷ Core thalamo-cortical — Layer 2/3 basal

$\mathbf{k}^h \leftarrow \mathbf{W}_K^h \mathbf{x}$ ▷ Matrix thalamo-cortical — Layer 2/3 apical

$\mathbf{q}^h \leftarrow \mathbf{W}_Q^h \mathbf{x}$ ▷ Matrix thalamo-cortical — Layer 5 IT basal

$\mathbf{L5ET}^h \leftarrow \text{zeros}(d_v)$

for $i = 1, \dots, d_v$ **do**

for $j = 1, \dots, d_k$ **do**

$\mathbf{L23}^h[i, j] \leftarrow \mathbf{L23}^h[i, j] + \frac{1}{\tau^h} (\mathbf{v}^h[i] \cdot \phi(\mathbf{k}^h[j]) - g_\ell \mathbf{L23}^h[i, j])$ ▷ eq. 4 — Layer 2/3 soma

$\mathbf{L5IT}^h[i, j] \leftarrow \phi(\mathbf{q}^h[j]) \cdot \mathbf{L23}^h[i, j]$ ▷ eq. 5 — Layer 5 IT soma

$\mathbf{L5ET}^h[i] \leftarrow \mathbf{L5ET}^h[i] + \mathbf{L5IT}^h[i, j]$ ▷ eq. 12 — Layer 5 ET soma

end for

for $k = 1, \dots, d$ **do**

$\mathbf{y}[k] \leftarrow \mathbf{y}[k] + \mathbf{W}_Y^h[k, i] \cdot \mathbf{L5ET}^h[i]$ ▷ Cortico-thalamic, eq. 3 — Higher-order nucleus

end for

end for

$\mathbf{e} = \mathbf{t} - \mathbf{y}$ ▷ Thalamic error computation

for $h = 1, \dots, n_h$ **do**

Learning $\mathbf{W}_Y$

for $k = 1, \dots, d$ **do**

for $i = 1, \dots, d_v$ **do**

$\mathbf{W}_Y^h[k, i] \leftarrow \mathbf{W}_Y^h[k, i] + \eta_Y \cdot \mathbf{e}[k] \cdot \mathbf{L5ET}^h[i]$ ▷ eq. 10 — Updating cortico-thalamic output synapses

end for

end for

Learning $\mathbf{W}_Q$

$\mathbf{b}^h = \mathbf{W}_Y^{h\top} \mathbf{e}$ ▷ Thalamo-cortical error backpropagation

$\mathbf{L6CT}^h \leftarrow \text{zeros}(d_k)$

for $j = 1, \dots, d_k$ **do**

for $i = 1, \dots, d_v$ **do**

$\mathbf{L6IT}^h[i, j] \leftarrow \mathbf{b}^h[i] \cdot \mathbf{L23}^h[i, j]$ ▷ eq. 12 — Layer 6 IT soma

$\mathbf{L6CT}^h[j] \leftarrow \mathbf{L6CT}^h[j] + \mathbf{L6IT}^h[i, j]$ ▷ eq. 13 — Layer 6 CT soma

end for

for $k = 1, \dots, d$ **do**

$\mathbf{W}_Q^h[j, k] \leftarrow \mathbf{W}_Q^h[j, k] + \eta_Q \cdot \mathbf{L6CT}^h[j] \cdot \mathbf{x}[k]$ ▷ eq. 11 — Updating thalamo-cortical query synapses

end for

end for

end for

Algorithm 3 Multihead linear self-attention in cortico-thalamic circuits — Inference & Learning

```

for first to last MHSA block do      # Inference in Multihead Self-Attention (MHSA) blocks
  for each head/area  $h$  do
    Calculate keys, values, and queries:                                ▷ Layer 2/3 and 5 — thalamo-cortical
       $\mathbf{k}_t = \mathbf{W}_K^h \mathbf{x}_t, \mathbf{v}_t = \mathbf{W}_V^h \mathbf{x}_t, \mathbf{q}_t = \mathbf{W}_Q^h \mathbf{x}_t$ 
    Update key-value memory (eq. 1):                                    ▷ Layer 2/3 — cortical recurrent
       $\mathbf{M}_t^h = \mathbf{M}_{t-1}^h + \frac{1}{\tau^h} (\mathbf{v}_t^h \phi(\mathbf{k}_t^h)^\top - g_\ell \mathbf{M}_{t-1}^h)$ 
    Output of cortical area  $h$  (eq. 3):                                ▷ Layer 5 — cortico-cortical
       $\mathbf{o}_t^h = \mathbf{M}_t^h \mathbf{A} \phi(\mathbf{q}_t^h)$                                 ▷  $\mathbf{A}$ =Attention matrix, sym. & pos.def, id for transformers
  end for Loop through areas/heads  $h$  for inference
   $\mathbf{y}_t = \sum_h \mathbf{W}_Y^h \mathbf{o}_t^h$                                        ▷ Combine all the heads
  if MHSA block is hidden then
     $\mathbf{x}_t = \mathbf{y}_t$                                                     ▷ Token for next MHSA block – cortico-thalamic
  else last MHSA block
     $\mathbf{x}_t = \mathbf{y}_t$                                                     ▷ Output from whole MHSA block sequence
    return  $\mathbf{y}_t$                                                        ▷ Prediction of next input token
  end if
end for Loop through MHSA blocks until last: inference

for last to first MHSA block do      # Defining errors and learning by gradient descent
  if MHSA block is last one then
    Token error = next input token – prediction (decoding error eq. 31):  ▷ Higher thalamic nucleus
     $\mathbf{e}_t = \mathbf{x}_{t+1} - \mathbf{y}_t$                                        ▷ Thalamo-thalamic
  else MHSA block is hidden
     $\mathbf{e}_t^+ =$  error from upper MHSA block                                ▷ Case  $\phi(x) = x$ 
     $\mathbf{e}_t = \sum_h \mathbf{W}_Q^{h\top} \mathbf{A} \mathbf{M}_t^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t^+$           ▷ Upper thalamic nucleus
     $\mathbf{e}_t = \sum_h \mathbf{W}_Q^{h\top} \mathbf{A} \mathbf{M}_t^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t^+$           ▷ Thalamo-thalamic backprop error
  end if
  for each head/area  $h$  do                                           ▷ Gradient-based synaptic plasticity
    Query error (encoding error):                                       ▷ Cortical area – thalamo-cortical backprop error
     $\mathbf{e}_t^{q,h} = \mathbf{A} \mathbf{M}_t^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t$                                 ▷ Case  $\phi(x) = x$ 
    Update encoding weights (eq. 10):  $\dot{\mathbf{W}}_Y^h \propto \mathbf{e}_t \mathbf{o}_t^{h\top}$                                 ▷ Thalamo-cortical backprop error
    Update decoding weights (eq. 11):  $\dot{\mathbf{W}}_Q^h \propto \mathbf{e}_t^{q,h} \mathbf{x}_t^\top$                                 ▷ Cortical area – Case  $\phi(x) = x$ 
  end for Loop through areas/heads  $h$  for learning
end for Loop through MHSA blocks for learning
  
```

Algorithm 4 An equivalent formulation for $\phi(x) = x$ (and $\mathbf{M}^h = \mathbf{W}_V^h \widetilde{\mathbf{M}}^h \mathbf{W}_K^{h\top}$) – Inference and Plasticity

Require: $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{x}_{\text{next}} \in \mathbb{R}^d$, $\widetilde{\mathbf{M}}^h \in \mathbb{R}^{d \times d}$, $\mathbf{W}_V^h \in \mathbb{R}^{d_v \times d}$, $\mathbf{W}_K^h \in \mathbb{R}^{d_k \times d}$, $\mathbf{W}_Q^h \in \mathbb{R}^{d_k \times d}$, $\mathbf{W}_Y^h \in \mathbb{R}^{d \times d_v}$

for first to last MHSa block **do** **# Inference in MHSa blocks**

$\mathbf{y} \leftarrow \mathbf{0}$ $\triangleright$ Initialize network output

for $h = 1, \dots, n_h$ **do**

$\widetilde{\mathbf{M}}^h \leftarrow \widetilde{\mathbf{M}}^h + \frac{1}{\tau_h}(\mathbf{x}\mathbf{x}^\top - g_\ell \widetilde{\mathbf{M}}^h)$ $\triangleright$ Token correlation trace

$\mathbf{y} \leftarrow \mathbf{y} + \mathbf{W}_Y^h \mathbf{W}_V^h \widetilde{\mathbf{M}}^h \mathbf{W}_K^{h\top} \mathbf{A} \mathbf{W}_Q^h \mathbf{x}$ $\triangleright$ Add output from current haead/area

end for $\triangleright (\mathbf{A}$ apical attention on L5, sym. and pos.-definite)

if MHSa block is hidden **then**

$\mathbf{x} \leftarrow \mathbf{y}$ $\triangleright$ Token for next MHSa block

else last MHSa block

return $\mathbf{y}$ $\triangleright$ Output from whole MHSa block sequence

end if $\triangleright$ Prediction of next input token

end for

for last to first MHSa block **do** **# Plasticity in MHSa blocks**

if last MHSa block **then**

$\mathbf{e} \leftarrow \mathbf{x}_{\text{next}} - \mathbf{y}$ $\triangleright$ Prediction error of next input token

else MHSa block is hidden

$\mathbf{e}_t^+ = \text{error from upper MHSa block}$

$\mathbf{e} \leftarrow \sum_h \mathbf{W}_Q^{h\top} \mathbf{W}_K^h \widetilde{\mathbf{M}}^h \mathbf{W}_V^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e}_t^+$ $\triangleright$ Backpropagated error across thalamic nuclei

end if

for $h = 1, \dots, n_h$ **do**

$\mathbf{W}_Y^h \leftarrow \mathbf{W}_Y^h + \eta \left[\mathbf{e} \mathbf{x}^\top \mathbf{W}_Q^{h\top} \mathbf{W}_K^h \widetilde{\mathbf{M}}^h \mathbf{W}_V^{h\top} \right]$ $\triangleright$ Adapting cortical output synapses

$\left(\mathbf{W}_V^h \leftarrow \mathbf{W}_V^h + \eta \left[\mathbf{W}_Y^{h\top} \mathbf{e} \mathbf{x}^\top \mathbf{W}_Q^{h\top} \mathbf{W}_K^h \widetilde{\mathbf{M}}^h \right] \right)$ $\triangleright$ Only if memory structure not fixed

$\left(\mathbf{W}_K^h \leftarrow \mathbf{W}_K^h + \eta \left[\mathbf{W}_Q^h \mathbf{x} \mathbf{e}^\top \mathbf{W}_Y^h \mathbf{W}_V^h \widetilde{\mathbf{M}}^h \right] \right)$ $\triangleright$ Only if memory structure not fixed

$\mathbf{W}_Q^h \leftarrow \mathbf{W}_Q^h + \eta \left[\mathbf{W}_K^h \widetilde{\mathbf{M}}^h \mathbf{W}_V^{h\top} \mathbf{W}_Y^{h\top} \mathbf{e} \mathbf{x}^\top \right]$ $\triangleright$ Adapting query encoding synapses

end for

end for

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